

# Exact Reductions for Two Open Irrationality Problems of Erdős

A Lean formalisation of Problem #249 and the half-value branch of Problem #257

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## Abstract

Let  $S = \sum_{n \geq 1} \varphi(n)/2^n$ , and for  $A \subseteq \mathbb{N}_{>0}$  let  $X_A = \sum_{n \in A} (2^n - 1)^{-1}$ . Erdős asked whether  $S$  is irrational (Problem #249) and whether  $X_A$  is irrational for every infinite  $A$  (Problem #257). Both problems remain open. We formalise in Lean two exact reductions that isolate the missing unbounded step in each question.

For #249, irrationality of  $S$  is equivalent to the existence, beyond every bound, of a finite integer calculation plus a rigorous tail estimate showing that two shifted binary tails differ by a non-integer. We also prove unconditionally that any rational representation  $S = p/q$  has  $q > Q_0 \approx 7.96 \times 10^{34}$ . At the lcm scales  $M_t = \text{lcm}(1, \dots, t)$ , Lean checks 28 such calculations through  $t = 64$ . These checks remain finite; they do not prove the required unbounded supply.

For #257, we formalise the classical irrationality theorem for the sum over all positive exponents in every integer base  $b \geq 2$ , together with several structured infinite choices of  $A$ . At base 2, each weight exceeds the sum of all later weights, so every attainable value has a unique support. The value  $1/2$  is attainable exactly when its greedy expansion omits weights at arbitrarily large ranks; this would refute the universal assertion in #257. Nonattainment is equivalent to a final omission. After a putative final omission, the next-step recurrence has two branches. We exclude one branch and the integer remainder  $-3$  in the other; the values  $-2, -1$  and a future-tail inequality remain unresolved.

An exact change of variables places both series in one common formula, but transfers no irrationality theorem between them. We prove neither the irrationality of  $S$  nor the universal assertion in #257, and we do not decide whether  $1/2$  is attainable.

## 1 Introduction

The two questions are whether  $S = \sum_{n \geq 1} \varphi(n)/2^n$  is irrational (#249), and whether  $\sum_{n \in A} (2^n - 1)^{-1}$  is irrational for every infinite  $A \subseteq \mathbb{N}_{>0}$  (#257). Here  $\varphi(n)$  is Euler's totient, the number of integers from 1 to  $n$  that are coprime to  $n$ . Problem #249 appears on p. 61 of Erdős–Graham and p. 102 of Erdős's survey [3, p. 61][4, p. 102]; problem #257 appears on pp. 62 and 105, respectively [3, p. 62] [4, p. 105]. Numbering follows Bloom's catalogue [23]. Neither problem is settled.

Throughout,  $\mathbb{N} = \{0, 1, 2, \dots\}$  and  $\mathbb{N}_{>0} = \{1, 2, 3, \dots\}$ . A *support*  $A \subseteq \mathbb{N}_{>0}$  selects terms of the #257 series, and its sum belongs to the corresponding *achievement set*  $\mathcal{A}$ . Exponent 0 is excluded because  $2^0 - 1 = 0$ . An infinite support with sum  $1/2$  would therefore refute the universal assertion in #257.

## What is proved

Each question is reduced to a finite Lean-checkable calculation and one unbounded condition that remains open. For #249, a *certificate* is a finite integer calculation plus a rigorous bound on the omitted tail. Put  $M_t = \text{lcm}(1, \dots, t)$  and  $R_N = \sum_{m \geq 1} \varphi(N + m)2^{-m}$ . For #257, skipping rank  $n$  means omitting the weight  $(2^n - 1)^{-1}$ . The two central equivalences, stated before the later proof machinery, are:

$$\begin{aligned} \text{finite certificates for } R_{2M_t} - R_{M_t} \notin \mathbb{Z} \\ \text{at arbitrarily large } t &\iff S \notin \mathbb{Q}, \\ \text{the greedy expansion of } \frac{1}{2} \text{ skips ranks} \\ \text{beyond every bound} &\iff \frac{1}{2} \in \mathcal{A}, \\ &\implies \text{a counterexample to universal \#257.} \end{aligned}$$

The first line is an exact reformulation of #249. The second concerns only the proposed half-value counterexample to #257, not the universal statement. Neither line proves the required unbounded behaviour. The current analysis of the final possible skip reduces the second open condition to two explicit obligations; neither obligation is proved.

The totient constant also has the elementary form

$$S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1),$$

where  $X, Y \in \mathbb{N}_{>0}$  are independent and  $\Pr(X = n) = \Pr(Y = n) = 2^{-n}$ . Both series are values of one Lambert transform, but this common description transfers no theorem from one problem to the other; Section 6 gives the exact map.

| Question or result                                | Status                 | What is proved                                                                  |
|---------------------------------------------------|------------------------|---------------------------------------------------------------------------------|
| Irrationality in #249                             | Open                   | No proof of $S \notin \mathbb{Q}$ is claimed.                                   |
| Denominator exclusion                             | Proved                 | If $S = p/q$ , then $q > Q_0$ .                                                 |
| Tail differences and $M_t$ -diagonal certificates | Exact equivalences     | The unbounded supply is equivalent to $S \notin \mathbb{Q}$ , but remains open. |
| Diagonal computation                              | Finite evidence        | Lean checks 28 diagonal certificates through $t = 64$ ; this is not cofinal.    |
| Universal assertion in #257                       | Open                   | No proof covers every infinite support.                                         |
| Full and structured supports                      | Classical/formalised   | Full support and several restricted families are proved irrational.             |
| Membership of $1/2$                               | Open; exact reductions | Greedy and carry laws give equivalent events without proving they occur.        |

**What “Lean-checked” means here.** A linked theorem statement has a proof term accepted by Lean from the pinned source and dependencies. An exact equivalence remains an equivalence even when both sides are open; a finite calculation remains finite; and a theorem with a rationality hypothesis does not prove that a rational support exists. The manuscript, its link checker, and its generated indexes help a reader find the proof but are not themselves proof authority. The release section states the additional source-pin and trust checks required for a reproducible copy.

**Reading map.** Section 2 states the two exact reductions, and Section 3 introduces their common coordinates. Sections 4 and 5 give the #257 and #249 results. Section 6 explains the shared transform and the limits of the comparison. Sections 7 and 8 give the verification protocol and remaining open problems. The appendices contain the local arguments and audit tables. Linked phrases open the corresponding Lean proofs.

**Keywords.** irrationality; Erdős–Borwein constant; Euler totient; Lambert series; achievement sets; binary carry systems; Lean 4. **MSC 2020.** 11J72 (primary); 11A25, 68V20 (secondary).

## 2 The two exact reductions

The first equivalence below restates #249 itself. The second restates only the proposed half-value counterexample to #257. Neither equivalence proves the unbounded behaviour that it identifies.

### 2.1 Erdős #249: tail differences and finite certificates

Write

$$R_N = \sum_{m \geq 1} \frac{\varphi(N + m)}{2^m}.$$

Each  $R_N$  is  $2^N S$  minus an integer. Rationality of  $S$  would force one binary period  $h$  whose tail differences  $R_{N+h} - R_N$  are eventually all integral; the certificates below refute exactly such candidate periods (Section 5.3).

**Exact reduction.** The formal source proves the exact chain

$$\begin{aligned} S \notin \mathbb{Q} &\iff R_{N+h} - R_N \notin \mathbb{Z} \text{ for every } h > 0 \text{ and every } N \\ &\iff \text{for every } h > 0, N \text{ there is an } L \text{ with } \text{Sep}(h, N, L). \end{aligned}$$

Here  $\text{Sep}(h, N, L)$  is a finite residue computation with a rigorous tail allowance. At fixed  $h, N$ , some depth succeeds exactly when the tail difference is non-integral; integrality of one positive-shift difference forces  $S$  to be rational. These statements are the [tail-difference equivalence](#), [pointwise certificate equivalence](#), and the fixed-parameter [certificate-completeness theorem](#).

**Unconditional finite baseline.** Independently of the unbounded certificate supply, if  $S = p/q$ , then

$$q > Q_0 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053.$$

This is the [denominator exclusion](#).

**Missing theorem.** An unbounded certificate supply is the exact open statement: for every positive  $h$ , non-integral differences  $R_{N+h} - R_N$  occur at arbitrarily large  $N$ . Theorem 5.7 gives the equivalent one-parameter lcm-diagonal form. The 28 changing-lcm scales through  $t = 64$  are verified, but neither this finite computation nor the denominator bound proves the unbounded supply.

## 2.2 Erdős #257: half-value membership and final greedy skips

Let

$$\mathcal{A} = \left\{ \sum_{n \in A} \frac{1}{2^n - 1} : A \subseteq \mathbb{N}_{>0} \right\}.$$

**Exact reduction and finite exclusion.** The weights satisfy  $w_n > \sum_{m>n} w_m$ , so a value has at most one support  $A \subseteq \mathbb{N}_{>0}$ . No finite such support can have value  $1/2$ . For  $1/2$ , the following are equivalent:

$$\begin{aligned} \frac{1}{2} \in \mathcal{A} &\iff \text{the set of exponents omitted by the greedy expansion is infinite} \\ &\iff \text{a successor terminal bit equal to zero occurs beyond every bound} \end{aligned}$$

(the terminal bit is defined in Section 4). Concretely, the greedy expansion of  $1/2$  skips rank 1, takes ranks 2, 3, skips 4, 5, and takes 6, 7; membership asks whether such skips recur beyond every bound.

**Counterexample consequence.** Membership would produce an infinite support of rational sum and refute the universal statement of Erdős #257. Thus the counterexample condition is exact: terminal zeros beyond every bound, equivalently no last greedy skip. The formal statements are the [infinite-skip equivalence](#), [unbounded-terminal equivalence](#), [no-last-skip equivalence](#), and the [finite-support exclusion](#).

**Missing theorem.** The missing input is a proof that the terminal bit is zero beyond every bound. The current last-skip route proves that, after a putative last skip, neither the upper transition nor middle carry  $-3$  can occur. It remains to exclude middle carries  $-2, -1$  and to prove a future-tail inequality for the remaining nonnegative middle carries. Neither statement is known for the greedy orbit. The exact conditional implication is the [remaining-tail implication](#). Proposition 4.15 gives the local exclusions and Corollary 4.16 states the sufficient global hypothesis.

**Relation to prior work.** Prior results are cited at their first use in the body, beginning in Section 3. No priority claim is made for the reductions above.

### 3 Lambert-series identities and comparison values

Whenever  $\sum_{n \geq 1} |f(n)|/(2^n - 1) < \infty$ , define the Lambert transform by

$$L(f) = \sum_{n \geq 1} \frac{f(n)}{2^n - 1} = \sum_{m \geq 1} \frac{(f * \mathbf{1})(m)}{2^m}, \quad (f * \mathbf{1})(m) = \sum_{d|m} f(d),$$

the second equality being the absolutely convergent Lambert rearrangement. It follows by expanding  $(2^n - 1)^{-1} = \sum_{k \geq 1} 2^{-nk}$ , interchanging the sums, and collecting the terms with  $m = nk$ . The coefficient  $(f * \mathbf{1})(m)$  is the Dirichlet convolution of  $f$  with the constant function  $\mathbf{1}(n) = 1$ . We also write  $\text{Id}(n) = n$ , let  $\mu$  denote the Möbius function, and put  $\sigma(m) = \sum_{d|m} d$ . Apostol gives the classical arithmetic-function and Dirichlet-convolution background [6]. Merca introduced a Lambert-series factorisation theorem [7], and Merca–Schmidt later unified the associated matrix formulations [8].

The rearrangement separates two operations: the divisor transform  $f \mapsto f * \mathbf{1}$ , followed by evaluation of the resulting coefficients at binary scales,  $\gamma \mapsto \sum_{m \geq 1} \gamma(m)2^{-m}$ . Section 6 uses this decomposition to compare the two open questions. Three inputs form the convolution chain

$$\alpha \xrightarrow{*1} \varphi \xrightarrow{*1} \text{Id}, \quad \alpha = \varphi * \mu,$$

whereas  $\mu$  and  $\mathbf{1}$  are comparison inputs. Write  $E = L(\mathbf{1}) = \sum_{n \geq 1} (2^n - 1)^{-1}$  for the Erdős–Borwein constant. The weight  $\alpha$  is nonnegative; for example,  $\alpha(p) = p - 2$  at a prime  $p$ . The corresponding values have the following statuses.

| input $f$                | value                               | mathematical status | treatment here                   |
|--------------------------|-------------------------------------|---------------------|----------------------------------|
| <i>Convolution chain</i> |                                     |                     |                                  |
| $\alpha = \varphi * \mu$ | $L(\alpha) = S$                     | open (#249)         | identities and bounds formalised |
| $\varphi$                | $L(\varphi) = 2$                    | rational            | formalised                       |
| $\text{Id}$              | $L(\text{Id}) = \sum \sigma(m)/2^m$ | transcendental      | cited [13]                       |
| <i>Comparison inputs</i> |                                     |                     |                                  |
| $\mu$                    | $L(\mu) = 1/2$                      | rational            | formalised                       |
| $\mathbf{1}$             | $L(\mathbf{1}) = E$                 | irrational          | formalised                       |

The arrows apply only to the first three rows. The rows for  $\mu$  and  $\mathbf{1}$  are separate comparisons, and the table records arithmetic statuses rather than implications between them. In particular,  $S$  and  $E$  have the same Mersenne-denominator form, with weights  $\alpha$  and  $\mathbf{1}$ , but no displayed convolution connects those weights. The known status of one row therefore does not transfer to another.

Nesterenko proved that  $\mathbb{Q}(q, E_2(q), E_4(q), E_6(q))$  has transcendence degree at least 3 when  $0 < |q| < 1$ . Since  $q = 1/2$  is algebraic, the three Eisenstein values at  $1/2$  are algebraically independent. In particular,

$$L(\text{Id}) = \sum_{m \geq 1} \frac{\sigma(m)}{2^m} = \frac{1 - E_2(1/2)}{24}$$

is transcendental [13].

**Proposition 3.1** (Two rational Lambert identities).

$$L(\mu) = \sum_{d \geq 1} \frac{\mu(d)}{2^d - 1} = \frac{1}{2}, \quad L(\varphi) = \sum_{d \geq 1} \frac{\varphi(d)}{2^d - 1} = 2.$$

These equalities are formalised as the [Möbius identity](#) and the [totient identity](#). Möbius inversion gives  $\mu * \mathbf{1} = (1, 0, 0, \dots)$ , while  $\varphi * \mathbf{1} = \text{Id}$ ; the rearranged sums are therefore  $2^{-1}$  and  $\sum_{m \geq 1} m 2^{-m} = 2$ , respectively.

**Proposition 3.2** (Lambert representation of  $S$ ). *With the weight  $\alpha = \varphi * \mu$ ,*

$$\sum_{d \geq 1} \frac{\alpha(d)}{2^d - 1} = \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = S.$$

The first equality is formalised as the [Lambert representation](#). The second is the definition of  $S$ . The proposition places  $S$  in the same  $\sum(\cdot)/(2^d - 1)$  family as  $E$ , with the nonnegative weight  $\alpha$ .

**Probabilistic reading.** If  $X, Y$  are independent fair-coin waiting times,  $\Pr(X = n) = 2^{-n}$ , then

$$S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1).$$

Thus Erdős #249 also asks whether the probability that two independent fair-coin waiting times are coprime is irrational. The exact identity is Proposition 5.2; its Möbius-square form and the periodic and finite-algebraic variants of these identities are collected in Appendices D.1–D.2.

**What this section proves—and does not prove.** The formalised contribution is  $L(\mu) = 1/2$ ,  $L(\varphi) = 2$ , and  $L(\alpha) = S$ ; the transcendental row is cited from Nesterenko. None of these statuses transfers between inputs. For #249 the remaining issue is still the unbounded certificate supply isolated in Section 5; for #257 this section supplies context rather than either the universal theorem or a rational counterexample.

## 4 Erdős–Borwein-type irrationality (the #257 direction)

We begin with the classical full-support theorem and named structured supports. We then specialise to base 2, where each weight exceeds the sum of all later weights and hence determines a unique support. In these coordinates we derive necessary conditions on a hypothetical rational value, then return to the target  $1/2$  and its final-skip reduction. None of these results covers arbitrary infinite supports, so the universal assertion in Erdős #257 remains open.

## 4.1 Full support: the Erdős–Borwein constant, every base

**Theorem 4.1** (full-support irrationality). *For every integer  $b \geq 2$ , the series  $\sum_{n \geq 1} \frac{1}{b^n - 1}$  is irrational. In particular the Erdős–Borwein constant  $E = \sum_{n \geq 1} 1/(2^n - 1)$  is irrational.*

The [all-base theorem](#) and its [base-2 corollary](#) are formalised. This is Erdős’s 1948 theorem [1]. In Section 3 it is the value  $L(1)$ , and it is the fixed machine-checked comparison point in the same Mersenne–Lambert family as the open  $S$ ; as Section 3 notes, it is not a further convolution step from  $S$ . Borwein later proved a wider rational-shift Lambert-series theorem [5]; that result is not formalised here.

The proof begins with the Lambert identity

$$\sum_{n \geq 1} \frac{1}{b^n - 1} = \sum_{m \geq 1} \frac{\tau(m)}{b^m},$$

where  $\tau(m)$  is the number of positive divisors of  $m$ ; see the [Lambert identity](#). If the right-hand side were rational with reduced denominator  $q$ , then a non-integral integer multiple could not lie at distance less than  $1/q$  from an integer. It is therefore enough, for every  $q$ , to construct  $N$  such that  $b^N \sum_m \tau(m)b^{-m}$  is non-integral but lies within  $1/q$  of an integer. This is the formal [near-integer criterion](#).

The construction of  $N$  is divided into three finite parts. First, a bounded use of Bertrand’s postulate supplies disjoint prime blocks, and the Chinese remainder theorem chooses an arithmetic progression on which

$$b^r \mid \tau(N + r) \quad (1 \leq r \leq K).$$

These divisibilities make the first  $K$  terms of the shifted tail integral. The multiplicativity step that converts prescribed prime valuations into the displayed divisibility is the [prime-block divisibility lemma](#). Second, divisor pairing bounds the average of the weighted middle window

$$\sum_{K < r \leq L} \tau(N + r)b^{L-r}$$

along that progression. A pigeonhole choice then selects one translate with a small middle contribution; see the [weighted-middle selection lemma](#). Third, the elementary estimate  $\tau(n) \leq n$  controls the infinite tail after  $L$ . The parameters are chosen so that the middle and far-tail bounds together are smaller than  $1/q$ ; the shifted tail is strictly positive because every divisor count is positive, so the dilated sum is not an integer.

Thus the analytic series is used only in the Lambert identity and the geometric tail estimate. The remaining proof is a finite CRT construction, divisor counting, and explicit parameter arithmetic. The final certificate existence theorem holds for every  $b \geq 2$  and every precision  $q$ ; see the [uniform certificate theorem](#). Its composition with the near-integer criterion is the theorem cited above. This is a formalisation of the classical full-support theorem, not a new irrationality result.

## 4.2 Named infinite-support cases

Beyond full support, the development proves irrationality for a family of infinite supports  $A$ , at every base  $b \geq 2$ . Each is a formalised case of the #257 statement family. None is the universal statement, and none is claimed as new mathematics.

**Theorem 4.2** (support-class family). *For every integer  $b \geq 2$ , the series  $\sum_{n \in A} 1/(b^n - 1)$  is irrational for each of the following infinite supports  $A$ :*

- (a) *factorial support  $A = \{n! : n \geq 1\}$ , giving  $\sum 1/(b^{n!} - 1)$ ;*

- (b) *power-of-two support*  $A = \{2^n : n \geq 0\}$ , giving  $\sum 1/(b^{2^n} - 1)$ ;
- (c) *multiples*  $A = d\mathbb{N}_{>0} = \{dk : k \geq 1\}$  with  $d \geq 1$ ;
- (d) *pairwise-coprime supports with summable reciprocals* (Erdős's condition [2]);
- (e) *eventually-periodic supports, residue classes, and the odd numbers*.

These sort into three groups. The classical case is (d): infinite pairwise-coprime  $A$  with  $\sum_{a \in A} 1/a < \infty$ , formalised from Erdős's condition. The lcm-gap named cases are (a) and (b), whose support gaps outgrow the running lcm; the structured families are (c) and (e).<sup>1</sup> Formalised case by case by the [factorial-support theorem](#), [power-of-two-support theorem](#), [multiple-support theorem](#), [pairwise-coprime-support theorem](#), and [eventually-periodic-support theorem](#). The residue-class and odd supports in part (e) are direct specialisations of the last theorem; the first two theorems already hold for every base  $b \geq 2$ .

The prime support is now known unconditionally: Tao and Teräväinen proved that

$$\sum_p \frac{1}{2^p - 1} = \sum_{n \geq 1} \frac{\omega(n)}{2^n}$$

is irrational [20]. Their proof uses quantitative two-point correlations of multiplicative functions and is not formalised here. This settles the prime-support case, while the universal support statement remains open.

Cases (a) and (b) both use the lcm-gap criterion. The power-of-two case is not a Liouville argument: before  $2^k$ , the prefix lcm is  $2^{k-1}$ , so the gap has only the same size as the lcm. Period noncollapse gives the finite input: the order of  $b$  modulo the reduced denominator of a partial sum equals that prefix lcm. The denominator therefore exceeds the lcm, while the additive gap to the next exponent still tends to infinity; the tail is then too small for a rational limit.

The results above enlarge the family of supports for which irrationality is known, but they do not supply the universal quantifier. We now fix the base at 2. The support sums form one achievement set, and an infinite-support counterexample is a rational point of this set whose unique support is infinite. We first derive necessary conditions for such a rational point and then specialise to  $1/2$ , for which finite support is impossible.

### 4.3 The base-2 achievement set

For  $n \geq 1$ , put

$$w_n = \frac{1}{2^n - 1}.$$

For  $n \geq 0$ , put

$$T_n = \sum_{m > n} w_m, \quad \mathcal{A} = \left\{ \sum_{n \in A} w_n : A \subseteq \mathbb{N}, 0 \notin A \right\}.$$

For  $x \geq 0$ , define the greedy residuals by  $r_0(x) = x$  and

$$\varepsilon_n(x) = \begin{cases} 1, & w_n \leq r_{n-1}(x), \\ 0, & w_n > r_{n-1}(x), \end{cases} \quad r_n(x) = r_{n-1}(x) - \varepsilon_n(x)w_n.$$

When  $\varepsilon_n(x) = 1$  we say the expansion *takes* rank  $n$ ; otherwise it *skips* rank  $n$ .

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<sup>1</sup>Luca–Tachiya prove the broader eventual-periodic rational-coefficient theorem; its indicator specialisation directly contains case (e) [15]. Their Chowla–Erdős-style large-modulus proof is distinct from the periodic-divisor certificate route formalised here, and the full mixed-sign coefficient theorem is not claimed.

**Theorem 4.3** (strict-tail greedy coding). *For every  $n \geq 1$ ,*

$$T_n < w_n.$$

*Hence every  $x \in \mathcal{A}$  has a unique support by positive exponents (the “normalised” support in the formal source). Moreover, a real number  $x$  lies in  $\mathcal{A}$  if and only if  $x \geq 0$  and  $r_n(x) \leq T_n$  for every  $n \geq 0$ . In that case  $x = \sum_{n \geq 1} \varepsilon_n(x) w_n$ .*

The strict-tail inequality and greedy criterion are the [strict-tail theorem](#) and [greedy membership theorem](#). Uniqueness is the [support-coding theorem](#). Kovač–Tao provide the strict-tail Cantor-set context [16].

**Geometric and finite-arithmetic consequences.** Proposition B.12 proves that  $\mathcal{A}$  is compact, perfect, totally disconnected, nowhere dense, and of Lebesgue measure 1. Proposition B.13 shows that the greedy recursion in exact rational arithmetic agrees step by step with the real recursion and gives sound finite certificates of nonmembership. For the half-value question, Theorem 4.3 makes the candidate support canonical, while compactness allows one to pass from cofinal finite approximations to a point of  $\mathcal{A}$ . The measure-one statement does not decide whether  $1/2 \in \mathcal{A}$ ; a finite certificate can prove nonmembership, but survival through any finite depth proves nothing about membership. These consequences use the same greedy coordinates but are not clauses of the membership equivalence. Their detailed statements and the one-sided certificate boundary are in Appendix B.5. No novelty claim is made for the strict-tail geometry. Before specialising to  $1/2$ , we record the carry constraints forced by any rational  $X_A \in \mathcal{A}$ ; these are necessary conditions in the universal direction, not yet a contradiction.

#### 4.4 Rigidity of a hypothetical rational support

For  $A \subseteq \mathbb{N}_{>0}$ , define

$$f_A(m) = \#\{a \in A : a \mid m\}, \quad X_A = \sum_{a \in A} \frac{1}{2^a - 1} = \sum_{m \geq 1} \frac{f_A(m)}{2^m}, \quad T_f(N) = \sum_{j \geq 1} \frac{f(N+j)}{2^j}.$$

Thus  $f_A(m)$  counts the selected exponents dividing  $m$ ,  $X_A$  is the original support sum, and  $T_f(N)$  is the binary tail of a coefficient sequence after position  $N$ .

**Theorem 4.4** (rational-support carry recurrence). *Suppose that  $A \neq \emptyset$  and*

$$X_A = \frac{p}{2^{c_v}}, \quad p \in \mathbb{Z}, \quad c \in \mathbb{N}, \quad v \geq 1 \text{ odd}.$$

*Then  $u_N = vT_{f_A}(c+N)$  is a positive integer for every  $N$ , and*

$$u_{N+1} + v f_A(c+N+1) = 2u_N. \tag{4.1}$$

This is formalised by [the shifted natural-state theorem](#). The recurrence is binary long division: since  $2T_f(M) = f(M+1) + T_f(M+1)$ , doubling the scaled tail separates the next coefficient, and rationality is what makes the states  $u_N$  integers. Its tail-orbit normal form is Theorem B.1. A two-element support, with its explicit period-six orbit and multiplier 21, is worked at the end of Appendix B.2.

**Corollary 4.5** (unbounded-state consequence). *Under the hypotheses of Theorem 4.4, if  $A$  is infinite, then  $(u_N)$  is unbounded.*

This is formalised by [the fraction-facing unbounded-state theorem](#); see also Theorem B.10.



**Corollary 4.6** (sublogarithmic coverage consequence). *Under the hypotheses of Theorem 4.4, for every  $\varepsilon > 0$  there is  $B = B(\varepsilon, c, v)$  such that every interval  $\{c + N + 1, \dots, c + N + \ell\}$  on which  $f_A$  vanishes satisfies*

$$\ell \leq \varepsilon \log_2(N + 1) + B;$$

This is formalised by [the sublogarithmic zero-window theorem](#); see also Theorem B.6.

**Corollary 4.7** (odd-denominator reciprocal-mass consequence). *Under the hypotheses of Theorem 4.4, if  $\rho(A) = \sum_{a \in A} a^{-1} < \infty$ ,  $v > 1$ , and  $h = \text{ord}_v(2)$ , then*

$$\rho(A) = \frac{w}{h} + \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{N < M} e_N \geq \frac{w}{h},$$

where  $w$  is the number of wraps in one doubling-residue cycle and  $e_N = \lfloor u_N/v \rfloor$  is the integral excess in the split  $u_N = ve_N + \bar{p}_N$ , with  $\bar{p}_N$  the least residue of  $p2^N$  modulo  $v$ . A step wraps precisely when doubling  $\bar{p}_N$  crosses  $v$ ; thus  $w$  counts the resulting carry digits over one cycle of length  $h$ . The term  $w/h$  is the cycle's mean residue contribution, while the displayed excess mean is nonnegative. If  $(p, v) = 1$ , then  $w \geq 1$ .

The exact excess-mean identity is [the shifted excess-mean theorem](#); see also Proposition B.8.

**Corollary 4.8** (dyadic reciprocal-mass consequence). *Under the hypotheses of Theorem 4.4, if  $v = 1$  and  $A$  is infinite, then either  $\rho(A)$  diverges or  $\rho(A) > 1$ .*

This is formalised by [the dyadic reciprocal-mass alternative](#); see also Corollary B.9.

Read together, the four statements say that rationality would force an unbounded integer state sequence for an infinite support, severely limit intervals containing no support divisor, and constrain the reciprocal mass according to whether the reduced denominator has an odd factor. They are independent necessary consequences of one rationality premise, not a contradiction. Appendices B.1–B.2 supply the common recurrence and Boolean certificate; Appendices B.3 and B.4 supply the independent filters. No corollary above appears as a hypothesis of another; their appendix proofs do share the carry machinery, in the order displayed in Appendix B.4.

**Corollary 4.9** (Boolean–Möbius orbit form of universal #257). *The universal assertion in Erdős #257 is equivalent to the following orbit statement: for every rational number  $p/q$ , every Boolean–Möbius carry certificate for  $p/q$  reconstructs a finite support. Equivalently, the universal assertion fails exactly when one such certificate reconstructs an infinite support.*

Concretely, a certificate is an integer sequence  $U$  with  $U(0) = p$ , positive states bounded by  $q(2\sqrt{N} + 4)$ , and  $q \mid 2U(N) - U(N + 1)$ . Dividing these differences by  $q$  gives an integer coefficient sequence  $f_U$ . Its Boolean Möbius condition is  $(\mu * f_U)(n) \in \{0, 1\}$ , and inversion reconstructs the support as  $A_U = \{n \geq 1 : (\mu * f_U)(n) = 1\}$ . These are exactly the conditions in Theorem B.4. That theorem is the Lean-checked fraction-by-fraction equivalence [between supports and carry certificates](#); the corollary merely quantifies it over  $p/q$  and separates finite from infinite reconstructed supports. This is an exact reformulation, not a finite algorithm: the orbit condition and the support it reconstructs are infinite objects.

## 4.5 The target $1/2$

**Why this target.** Any infinite support with any rational value would refute the universal statement. Fixing  $1/2$  turns one possible counterexample route into the point-membership question  $1/2 \in \mathcal{A}$ . This target is convenient, not forced.

The reason it is distinguished is the signed Möbius identity  $L(\mu) = 1/2$  from Section 3. A support series is different: each term has coefficient 0 or 1. The most immediate conversion, selecting exactly the indices with  $\mu(d) = -1$ , overshoots  $1/2$  by at least  $1/63$  (Proposition B.2).

Moreover, no finite support attains  $1/2$  (Proposition 4.12 below), while the strict-tail coding of Theorem 4.3 makes any candidate support unique.

Let

$$\mathcal{S}(x) = \{n \geq 1 : \varepsilon_n(x) = 0\}$$

be the set of exponents omitted by the greedy expansion.

**A worked start.** The greedy expansion of  $1/2$  begins:  $w_1 = 1 > \frac{1}{2}$ , so rank 1 is skipped;  $w_2 = \frac{1}{3}$  is taken, leaving  $\frac{1}{6}$ ;  $w_3 = \frac{1}{7}$  is taken, leaving  $\frac{1}{42}$ ; ranks 4, 5 are skipped because  $w_4 = \frac{1}{15}$  and  $w_5 = \frac{1}{31}$  exceed  $\frac{1}{42}$ ; and  $w_6 = \frac{1}{63}$ ,  $w_7 = \frac{1}{127}$  are taken, leaving

$$\frac{1}{2} - \left( \frac{1}{3} + \frac{1}{7} + \frac{1}{63} + \frac{1}{127} \right) = \frac{1}{16002}.$$

The exact calculation continues with the next take at rank 14, and through rank 26 it takes exactly

$$2, 3, 6, 7, 14, 20, 21, 26.$$

The finite seam model introduced below reproduces this prefix uniformly for every row  $s \geq 27$ ; this is the [kernel-checked rank-\(26\) prefix](#) used again in Appendix A.4.

This is only an initial segment. Membership of  $1/2$  in  $\mathcal{A}$  requires the complementary skips to recur beyond every bound. Nonmembership has a finite alternative: one skip is the *last*, precisely when its residual lies strictly above the whole remaining tail (Appendix A.4.6). Theorem 4.14 below identifies nonmembership with the existence of such a skip.

The next definition replaces the real greedy calculation by an integer one. A row  $s$  keeps the ranks  $2, \dots, s-1$ , rescales every weight by  $4^s$ , rounds down, and uses a target  $2^s$  below the rescaled value  $4^s/2$ . The resulting zero-one word records the finite take/skip decisions; its last entry is the terminal bit used in Theorem 4.11.

**Definition 4.10** (finite greedy seam and terminal bit). For an integer  $s \geq 6$ , set

$$q_{s,d} = \lfloor 4^s w_d \rfloor = \left\lfloor \frac{4^s}{2^d - 1} \right\rfloor \quad (2 \leq d < s), \quad K_s = 4^s \cdot \frac{1}{2} - 2^s = 2^{2s-1} - 2^s.$$

Starting from  $R_{s,2} = K_s$ , define successively, for  $2 \leq d < s$ ,

$$\beta_{s,d} = \begin{cases} 1, & q_{s,d} \leq R_{s,d}, \\ 0, & q_{s,d} > R_{s,d}, \end{cases} \quad R_{s,d+1} = R_{s,d} - \beta_{s,d} q_{s,d}.$$

The Boolean word  $(\beta_{s,2}, \dots, \beta_{s,s-1})$  is the *finite greedy seam at row  $s$* , and  $\tau_s = \beta_{s,s-1}$  is its terminal bit.

For example, at  $s = 7$  we have  $K_7 = 8064$ , and the run takes  $q_{7,2} = 5461$  and  $q_{7,3} = 2340$ , skips 1092 and 528, and takes  $q_{7,6} = 260$ , leaving 3: the word is 1, 1, 0, 0, 1, matching the real greedy decisions at ranks 2–6, with terminal bit  $\tau_7 = 1$ . The seam is thus an exact integer coordinate model of the real greedy process. Appendix A.4 develops the equivalence with final skips and the unresolved transition cells; the finite coordinate changes in Appendix A.5 do not strengthen that equivalence.

**Theorem 4.11** (exact half-membership classification). *The following are equivalent:*

- (i)  $\frac{1}{2} \in \mathcal{A}$ ;
- (ii)  $\mathcal{S}(1/2)$  is infinite;
- (iii)  $\mathcal{S}(1/2)$  has no largest element;
- (iv) for every  $N$  there is  $s \geq \max\{N, 6\}$  with  $\tau_s = 0$ .

The omitted-set equivalence is the [infinite-skip theorem](#); the terminal forms are the [unbounded-terminal theorem](#) and [no-final-skip theorem](#).

**Proposition 4.12** (finite half-value exclusion). *No finite support  $A \subseteq \mathbb{N}_{>0}$  has value  $1/2$ .*

This is the [finite-support theorem](#).

**Corollary 4.13** (counterexample from half-membership). *Any one of the equivalent conditions in Theorem 4.11 produces an infinite support  $A \subseteq \mathbb{N}_{>0}$  with  $X_A = 1/2$ , and therefore refutes the universal form of Erdős #257.*

This combines the unique greedy support from Theorem 4.3 with Proposition 4.12. A tempting signed-to-Boolean shortcut also fails: selecting precisely the negative Möbius indices overshoots  $1/2$  by at least  $1/63$ . Proposition B.2 gives the exact sign-separation identity.

Theorem 4.11 therefore leaves one question: does the greedy orbit of  $1/2$  skip cofinally? It does not answer that question. The next subsection characterises nonmembership by a final skip and identifies what would be required to exclude it.

## 4.6 The final-skip reduction

Appendix A.4 contains the detailed derivations used in this subsection; the statements and their logical status are recorded here. To detect whether a skip can be final, we compare the ranks selected in consecutive seam rows.

For each  $s \geq 6$ , let

$$G_s = \{d : 2 \leq d < s, \beta_{s,d} = 1\}.$$

Thus  $G_s$  is the set chosen by the integer greedy run at row  $s$ . Among all subsets of  $\{2, \dots, s-1\}$  whose integer weight exceeds  $K_s$ , let  $H_s$  be the unique one with the smallest weight. It is the cheapest strict overshoot of the target.

The recurrence from row  $s$  to row  $s+1$  has three forms:

$$\begin{aligned} \text{upper: } & G_{s+1} = H_s, \\ \text{middle: } & G_{s+1} = G_s, \\ \text{right: } & G_{s+1} = G_s \cup \{s\}. \end{aligned}$$

An upper step resets the selected ranks to  $H_s$ . A middle step keeps  $G_s$  and skips the new rank  $s$ ; a right step keeps  $G_s$  and takes that rank. A final skipped exponent  $D$  is therefore an upper or middle transition followed only by right transitions. The terminal bit at row 13 is false. Consequently, if the seam is eventually right, its last false terminal row satisfies  $D \geq 13$ ; the lower bound comes from this calculation rather than an omitted range of cases. This is the [last-false row theorem](#).

At a middle transition  $D \geq 13$ , the next question is whether restoring the skipped rank  $D$  still leaves the support below  $1/2$ . Let

$$A_D = G_D \cup \{D\}$$

be this augmented support. The two local divisor counts entering the next carry are packaged as

$$P_D = 2f_{G_D}(2D+1) + f_{G_D}(2D+2), \quad C_D = 4R_{D,D} - P_D - 4 \in \mathbb{Z}.$$

Thus  $C_D$  is the finite integer transition carry. The contribution from all later divisor counts is the infinite tail

$$\Theta_D = \sum_{j \geq 1} \frac{f_{A_D}(2D+2+j)}{2^j}.$$

This tail is complete, not a finite truncation. The residual identity of Appendix A.4.7 compares the two quantities exactly:

$$\frac{1}{2} - \sum_{a \in A_D} w_a = 2^{-(2D+2)}(C_D - \Theta_D),$$

so  $\Theta_D < C_D$  is equivalent to  $\sum_{a \in A_D} w_a < \frac{1}{2}$ .

For a finite set  $u \subseteq \{1, \dots, d\}$ , write  $V(u) = \sum_{n \in u} w_n$ . Once this prefix through rank  $d$  is fixed, the next decision is whether to take  $w_{d+1}$ . If it is skipped, even taking every later weight reaches at most  $V(u) + T_{d+1}$ ; if it is taken, the resulting sum is at least  $V(u) + w_{d+1}$ . We call  $(u, d)$  a *fatal half-gap* when  $1/2$  lies strictly between these two bounds:

$$V(u) + T_{d+1} < \frac{1}{2} < V(u) + w_{d+1}.$$

No support agreeing with  $u$  through rank  $d$  can then represent  $1/2$ : extensions omitting rank  $d+1$  fall short, while those containing it overshoot.

**Theorem 4.14** (final-skip classification). *Nonmembership of  $1/2$  in  $\mathcal{A}$  is equivalent to each of the following:*

- (i) *the integer seam recurrence is eventually right;*
- (ii) *a fatal half-gap exists;*
- (iii) *the real greedy expansion of  $1/2$  has a final skipped exponent.*

The three equivalent nonmembership criteria are formalised in [fatal-gap classification](#), [nonmembership classification](#), and [final-skip classification](#).

**Proposition 4.15** (excluded final-skip branches). *At a final skipped exponent  $D \geq 13$ , the upper successor is impossible. On the middle branch, the only carry values outside the ranges  $C_D \leq -4$  and  $C_D \geq 0$  are  $C_D \in \{-3, -2, -1\}$ . At a final skip, the value  $C_D = -3$  is impossible; the values  $-2$  and  $-1$  remain unresolved.*

The upper and  $C_D = -3$  exclusions are the [upper-branch exclusion](#) and [middle- \$-3\$  exclusion](#); the upper case is proved directly at the linked declaration, and Appendix A.4 sketches a prose argument only for the  $-3$  cell. The three exceptional values are identified by the [three-cell classification](#).

**Corollary 4.16** (tail-dominance membership criterion). *If every middle transition at a row  $D \geq 13$  with  $C_D \neq -3$  satisfies*

$$\Theta_D < C_D, \tag{4.2}$$

*then no final skipped exponent exists, and hence  $1/2 \in \mathcal{A}$ .*

This implication is the [tail-dominance theorem](#). Since  $\Theta_D \geq 0$ , inequality (4.2) cannot hold when  $C_D = -2$  or  $-1$ . The corollary therefore has two open inputs. First, these two carry values must be shown not to occur at any actual middle transition. Second, every remaining middle transition with  $C_D \geq 0$  must have complete future tail smaller than  $C_D$ . Neither input is proved for the greedy orbit. The word “every” is essential: one final middle transition is enough for nonmembership, so a bound that failed at even one row would not rule it out.

**Logical status of the final-skip argument.** The proved levels of the argument are

$$\begin{array}{lll} \frac{1}{2} \notin \mathcal{A} & \iff \text{a final skipped exponent exists} & [\text{proved}], \\ \text{final upper, or final middle with } C_D = -3 & \implies \perp & [\text{proved}]. \end{array}$$

These lines do not imply membership. The unproved hypothesis has two parts: exclude all actual middle transitions with  $C_D \in \{-2, -1\}$ , and prove (4.2) at every remaining non-(-3) middle transition. Together they establish the global hypothesis of Corollary 4.16; only then does Corollary 4.13 produce an infinite rational-valued support and a counterexample to the universal #257 statement.

## 4.7 What remains open

The two directions remain separate.

- (a) The universal problem is to prove  $X_A \notin \mathbb{Q}$  for every infinite  $A \subseteq \mathbb{N}_{>0}$ . Theorem 4.2 proves several families, while Theorem 4.4 and its four corollaries constrain a hypothetical rational support. Neither route supplies the universal quantifier.
- (b) The distinguished half-value route asks whether  $1/2 \in \mathcal{A}$ . Theorem 4.11 identifies this with infinitely many greedy skips, while Theorem 4.14 identifies nonmembership with a final skip. Proposition 4.15 proves that this final transition cannot be upper and that a final middle transition cannot have  $C_D = -3$ . Corollary 4.16 would prove membership if no actual middle transition had  $C_D = -2$  or  $-1$ , and if every remaining middle transition with  $C_D \geq 0$  satisfied the future-tail inequality (4.2). Neither statement is proved.

Appendix A records the exact local reductions, conditional compactness criteria, finite certificates, counterexamples to stronger local assertions, and arguments that rule out several proposed strategies. These results narrow the remaining cases; they decide neither open #257 question.

## 5 The totient constant $S$ (Erdős #249)

Whether

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

is irrational remains open. The argument has four logically distinct layers. First, a finite Farey computation excludes every rational denominator up to an explicit bound. Second, irrationality is characterised exactly by the existence of finite certificates against every possible eventual binary period. Third, many such certificates are verified at bounded parameters. Fourth, and still missing, is a theorem supplying them at unbounded parameters. Only the fourth layer would settle the problem.

The main section states this reduction. Appendix C.1 gives equivalent forms of the missing assertion that certificates occur at arbitrarily large scales. Appendix C.2 collects finite estimates and checked certificates; any conclusion of irrationality there assumes, rather than proves, an unbounded supply. Appendix C.3 records candidate sufficient conditions and shows why several natural shortcuts fail. Appendix C.4 proves exact dyadic rank growth but not the missing implication from that growth to irrationality. Only the first appendix is an exact reformulation of the open fourth layer.

## 5.1 Unconditional: no small denominator

**Theorem 5.1** (denominator exclusion). *Set*

$$Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}.$$

For every integer  $p$  and every  $q$  with  $1 \leq q \leq Q_0$ ,

$$S \neq \frac{p}{q}.$$

Equivalently, if  $S$  is rational then its reduced denominator exceeds that bound.

Formalised as the [record denominator exclusion](#). For a fixed window length  $K$ , the checked prefix and a rigorous tail bound define a finite gap certificate. If a denominator  $q$  failed that certificate, elementary rearrangement would put some rational  $r/q$  in a specific short interval. The larger bounds bracket that interval by fractions  $a/b < c/d$  satisfying  $bc - ad = 1$ . The mediant lemma says that every rational strictly between such neighbours has denominator at least  $b + d$ . Thus no  $q < b + d$  can fail the certificate; for each such  $q$ , the corresponding scaled tail lies strictly between consecutive integers, whereas  $S = m/q$  would force it to be an integer.

The finite exclusion ladder records the successive verified thresholds

$$4838 \rightarrow 4\,194\,304 = 2^{22} \rightarrow 2.49 \times 10^{17} \text{ (} K=120 \text{)} \rightarrow 7.96 \times 10^{34} \text{ (} K=240 \text{)}.$$

The mediant framework is historically associated with Farey's 1816 note on vulgar fractions [10]; every exclusion above is Lean-checked in the formal-source checkpoint. The first threshold is a direct low-carry check,  $2^{22}$  is retained as a named intermediate corollary, and the two largest thresholds use the Farey separation. See [denominator exclusion through 4838](#) and [the intermediate 4194304 bound](#), then [the 120-term Farey bound](#) and [240-term Farey bound](#) for the two Farey windows.

This is an explicit lower bound on the denominator of any rational representation, not a proof of irrationality. A rational with a larger denominator is not excluded by it.

*Remark* (one bound, four representations). Because the Lambert identities of Section 3 are equalities, the same finite record transfers verbatim to three other representations of the constant: to the Lambert representation  $\sum_d \alpha(d)/(2^d - 1)$  ([Lambert-representation denominator transfer](#)), to the signed Möbius-square constant  $T = \sum_d \mu(d)/(2^d - 1)^2$  at half the bound ([signed-square denominator transfer](#)), and to the coprimality probability of Section 5.2 at half the bound. Half appears because  $S = \frac{1}{2} + T$  turns a denominator  $d$  for  $T$  into  $2d$  for  $S$ .

## 5.2 The geometric picture: $S$ as coprime-pair mass

Let  $X, Y$  be independent fair-coin waiting times,  $\Pr(X = n) = 2^{-n}$  for  $n \geq 1$ . Then  $\Pr(d \mid X) = 1/(2^d - 1)$ , so the Lambert transform is a divisor calculus of  $X$ :  $L(f) = \mathbb{E}[(f * \mathbf{1})(X)]$ . The identities give  $L(\mu) = \Pr(X = 1) = 1/2$ ,  $L(\varphi) = \mathbb{E}[X] = 2$ ,  $L(\mathbf{1}) = \mathbb{E}[\tau(X)] = E$ , and  $L(\alpha) = \mathbb{E}[\varphi(X)] = S$ .

Counting visible lattice points  $(a, b)$  with  $a + b = n$ ,  $a \geq 1$ ,  $b \geq 0$ , and  $\gcd(a, b) = 1$  gives exactly  $\varphi(n)$ , uniformly in  $n$ , so  $S$  is itself a coprime-pair mass.

**Proposition 5.2** (coprimality probability).

$$S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1) = \frac{1}{2} + \sum_{\substack{a, b \geq 1 \\ \gcd(a, b) = 1}} 2^{-(a+b)}.$$

Formalised: [coprime-pair representation](#), with the underlying lattice identity at [visible-lattice-point identity](#). Base 2 is the one point where  $\Pr(X = a)\Pr(Y = b) = 2^{-(a+b)}$  needs no normalising constant. Restricting the half-open lattice mass to positive waiting times removes  $(1, 0)$ , producing the boundary term  $1/2$ . Thus #249 asks whether two independent fair-coin waiting times are coprime with irrational probability. After reducing  $(X, Y)$  by their gcd, the same distribution has an exact Stern–Brocot cylinder decomposition. Appendix D.3 states the resulting probability law and the sharp Fibonacci stability of its alternating runs. These are unconditional structural theorems inside one representation of  $S$ ; they do not supply the denominator survival needed for irrationality.

### 5.3 Exact certificate characterisation

The reduction uses only the elementary bound  $\varphi(n) \leq n$ , geometric tail estimates, and finite integer arithmetic. For  $t \geq 1$ , write  $M_t = \text{lcm}(1, \dots, t)$ .

**Why tail differences detect rationality.** Since  $2^N S = \sum_{n \leq N} \varphi(n) 2^{N-n} + R_N$ , each scaled tail  $R_N$  is  $2^N S$  minus an integer. Suppose  $S = p/q$  with  $q = 2^c v$  and  $v$  odd. For every multiple  $h$  of the multiplicative order of 2 modulo  $v$  and every  $N \geq c$ ,

$$R_{N+h} - R_N = \frac{2^N (2^h - 1)p}{q} - k \in \mathbb{Z}$$

for some integer  $k$ , because  $v \mid 2^h - 1$  and  $2^c \mid 2^N$ . Rationality therefore supplies one period  $h$  all of whose tail differences at large  $N$  are integral; conversely, a single integral positive-shift difference already forces  $S$  to be rational. A finite certificate refutes one candidate integrality, and refuting every period at arbitrarily large  $N$  refutes rationality outright. This is the mechanism behind the equivalences of this subsection.

**Definition 5.3** (tail differences and finite certificates). For  $N \geq 0$ , define the scaled tail

$$R_N = \sum_{m \geq 1} \frac{\varphi(N+m)}{2^m}.$$

For  $h \in \mathbb{N}_{>0}$  and  $N, L \in \mathbb{N}$ , define

$$\Delta_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j)) 2^{L-1-j} \in \mathbb{Z},$$

with the sum empty when  $L = 0$ , and say that  $(h, N, L)$  is a *finite certificate* when

$$\text{Sep}(h, N, L) : \quad N + h + L + 2 < (\Delta_{h,N,L} \bmod 2^L) < 2^L - (N + h + L + 2).$$

Here  $x \bmod 2^L$  denotes the least residue in  $\{0, \dots, 2^L - 1\}$ . The integer  $\Delta_{h,N,L}$  contains the first  $L$  binary places of  $R_{N+h} - R_N$ ; the two margins bound the omitted tail using  $\varphi(n) \leq n$ . Thus each instance of the predicate is decidable by finite integer arithmetic, while its conclusion concerns the infinite tail difference. The strict inequalities are impossible when  $L = 0$ , so every successful certificate has positive depth. Heuristically, the two margins together occupy a proportion of about  $2(N+h+L+2)/2^L$  of the modulus, which is small as soon as  $2^L$  is large compared with  $N + h + L$ ; a certificate can fail only when the residue lies exceptionally close to 0 or  $2^L$ , that is, when the tail difference is exceptionally close to an integer. Nothing in this paper converts that observation into the required supply at arbitrarily large  $N$ .

**Proposition 5.4** (certificate completeness). *For every  $h \geq 1$  and  $N \geq 0$ ,*

$$(\exists L, \text{Sep}(h, N, L)) \iff R_{N+h} - R_N \notin \mathbb{Z}.$$



This is the [certificate-completeness equivalence](#). Thus the finite predicate loses no information at a fixed pair  $(h, N)$ . Searching  $L = 1, 2, \dots$  is therefore a complete positive semi-decision procedure for non-integrality at fixed  $(h, N)$ : it halts exactly when the difference is non-integral. When the difference is integral, completeness says that no depth succeeds, so this search does not halt. It is not a two-sided decision procedure for an infinite tail.

**Theorem 5.5** (pointwise certificate characterisation). *The following are equivalent:*

$$\begin{aligned} S &\notin \mathbb{Q}, \\ \forall h \geq 1 \ \forall N \geq 0, \quad R_{N+h} - R_N &\notin \mathbb{Z}, \\ \forall h \geq 1 \ \forall N \geq 0 \ \exists L, \quad \text{Sep}(h, N, L). \end{aligned}$$

Formalised: [all-differences equivalence](#), and [pointwise certificate equivalence](#). The converse direction uses the fact that integrality of even one positive-shift tail difference forces  $S$  to be rational ([integral-difference rationality criterion](#)).

Here  $h$  tests a candidate rational period,  $N_0$  means “after every cutoff”,  $N \geq N_0$  chooses the location, and  $L$  is the finite binary depth checked there. A bounded list cannot cover every  $h$  and every  $N_0$ .

**Theorem 5.6** (exact tail-period characterisation).  *$S$  is irrational if and only if, for every period  $h \geq 1$  and every threshold  $N_0$ , there exist  $N \geq N_0$  and  $L$  with  $\text{Sep}(h, N, L)$ .*

Formalised as the [certificate-supply equivalence](#).

Every multiple of an eventual period is again a period, and every fixed  $h$  divides  $M_t$  for all  $t \geq h$ . The least common multiple therefore provides an exact one-parameter form: the single diagonal family  $(M_t, M_t)$  tests every candidate period at once.

**Theorem 5.7** (exact diagonal characterisation). *The following are equivalent:*

$$\begin{aligned} S &\notin \mathbb{Q}; \\ \forall t_0 \in \mathbb{N} \ \exists t \geq t_0 \ \exists L, \quad \text{Sep}(M_t, M_t, L). \end{aligned}$$

Formalised as the [exact lcm-diagonal characterisation](#). This replaces the two free parameters  $h, N$  by  $(M_t, M_t)$ ; producing such certificates at arbitrarily large  $t$  is still open.

**Example 5.8** (verified finite range). Lean checks:

- $\text{Sep}(h, 12, 16)$  for every  $1 \leq h \leq 8$ ;
- a fixed-window certificate for every  $1 \leq h \leq 16$ ;
- diagonal certificates  $\text{Sep}(M_t, M_t, L)$  at  $t = 1$  and every lcm-height change endpoint  $2 \leq t \leq 64$  (that is,  $M_{t-1} < M_t$ ), namely
$$t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59, 61, 64\};$$
- three finite menus of tails that cannot all share one fractional part.

These four classes are formalised by the [small-window certificate family](#), [periods through sixteen](#), [28 imported diagonal certificates through scale 64](#), and [cone-cell certificate table](#).

For example, at  $(h, N, L) = (1, 12, 16)$ ,

$$\Delta_{1,12,16} = -143140, \quad \Delta_{1,12,16} \bmod 2^{16} = 53468,$$

while the omitted-tail allowance is  $12 + 1 + 16 + 2 = 31$ . Hence

$$31 < 53468 < 2^{16} - 31,$$

which is exactly  $\text{Sep}(1, 12, 16)$  and therefore proves  $R_{13} - R_{12} \notin \mathbb{Z}$ . Pairwise rows name a non-integral difference; joint rows prove that one exists within the menu. They do not form an unbounded family.



## 5.4 What remains open

By Theorem 5.6, the unresolved statement is precisely

$$\forall h \geq 1 \ \forall N_0 \geq 0 \ \exists N \geq N_0 \ \exists L \geq 1, \quad \text{Sep}(h, N, L). \quad (5.1)$$

The diagonal condition in Theorem 5.7 is an equivalent one-parameter form of this statement. The verified scales in Example 5.8 form a finite set; they do not give certificates for every period after every cutoff. No theorem in the formal source proves (5.1); consequently no irrationality claim for  $S$  is made here.

*Remark* (finite inspection cannot establish the supply). The finite status of the verified scales is forced, not cautious. Fix a bound  $B \geq 1$  and a period  $P > B$ , and define  $\gamma(n) = \varphi(n)$  for  $n \leq B$ , while for  $n > B$  let  $\gamma(n) = n - 1$  if  $P \mid n$  and  $\gamma(n) = n$  otherwise. Then  $0 \leq \gamma(n) \leq n$ , and

$$\sum_{n \geq 1} \frac{\gamma(n)}{2^n} = \sum_{n \leq B} \frac{\varphi(n)}{2^n} + \frac{B+2}{2^B} - \frac{1}{2^P - 1}$$

is rational, and its reduced denominator has odd part exactly  $2^P - 1$ . Every finite computation that inspects only coefficients with index at most  $B$  — in particular every certificate  $\text{Sep}(h, N, L)$  with  $N + h + L \leq B$  — returns the same result for  $\gamma$  as for  $\varphi$ , and choosing  $P$  with  $2^P - 1 > Q_0$  makes the splice consistent with the denominator exclusion as well. This elementary splice is not formalised in the source and proves nothing about  $\varphi$  itself; it shows that a proof of (5.1) must use totient values at unbounded indices, so no extension of the finite record in Example 5.8 can establish the supply.

Appendix C.1 records the exact reformulations of this open obligation. Appendix C.2 records fixed-scale enclosures, projections, and certificates, followed by conditional implications that assume them at arbitrarily large scales. Appendices C.3–C.4 delimit candidate unbounded routes. None proves (5.1).

## 6 One transform and the limit of finite evidence

This section compares the structure of the two open reductions and the point where finite evidence stops. Every mathematical assertion in it is proved elsewhere in the paper; the section adds organisation, not theorems, and its closing remark is an elementary unformalised observation, flagged as such.

### 6.1 One transform, one mechanism, three questions

The Lambert rearrangement of Section 3 factors the common transform through two maps, the divisor transform followed by binary evaluation:

$$f \mapsto f * \mathbf{1} \mapsto \sum_{m \geq 1} \frac{(f * \mathbf{1})(m)}{2^m} = L(f).$$

Every series in this paper is an instance. The primitive-conductor input  $f = \alpha$  gives  $L(\alpha) = S$  with divisor stream  $\alpha * \mathbf{1} = \varphi$  (Proposition 3.2); a support indicator  $f = \mathbf{1}_A$  gives  $L(\mathbf{1}_A) = X_A$  with divisor stream  $f_A$  (Appendix B.2); and the comparison inputs  $\mu$ ,  $\mathbf{1}$ , and  $\varphi$  give the comparison values listed in Section 3.

Rationality acts on all of these through one mechanism. For any  $\mathbb{N}$ -valued stream  $\gamma : \mathbb{N} \rightarrow \mathbb{N}$  with  $\gamma(n) \leq n$ , a class containing both  $\varphi$  and every  $f_A$ , Theorem B.1 identifies rationality of  $\sum_n \gamma(n) 2^{-n}$  with the existence of a tempered integer orbit: after multiplication by one denominator, the coefficient stream is the exact one-step doubling difference  $2u(N) - u(N+1)$

of an integer sequence of size  $o(2^N)$ . The tail differences of Section 5.3 and the carry states of Theorem 4.4 are two coordinate systems for this one orbit condition. Möbius inversion then identifies the arithmetic input beneath the coefficient stream: the fixed weight  $\alpha = \varphi * \mu$  on the #249 side, and the Boolean indicator  $\mu * f_A = \mathbf{1}_A$  on the #257 side (Proposition B.3).

The three open questions are then three different questions about this one map.

- (i) Erdős #249 fixes the input and asks about one image point: is  $L(\alpha) = S$  rational?
- (ii) Universal #257 restricts the domain to indicators of infinite supports and asks whether the image of that domain meets  $\mathbb{Q}$  at all.
- (iii) The half-value route fixes the target and asks whether the fibre over  $1/2$  meets the same domain; by Theorem 4.3 the fibre over any value contains at most one positive-exponent support, so the question concerns one canonical greedy orbit.

Figure 1 displays the five inputs and the three question slices. The trichotomy describes where the questions sit inside one transform. It is not a theorem transferring information between them, and the next subsection records exactly why no transfer follows.

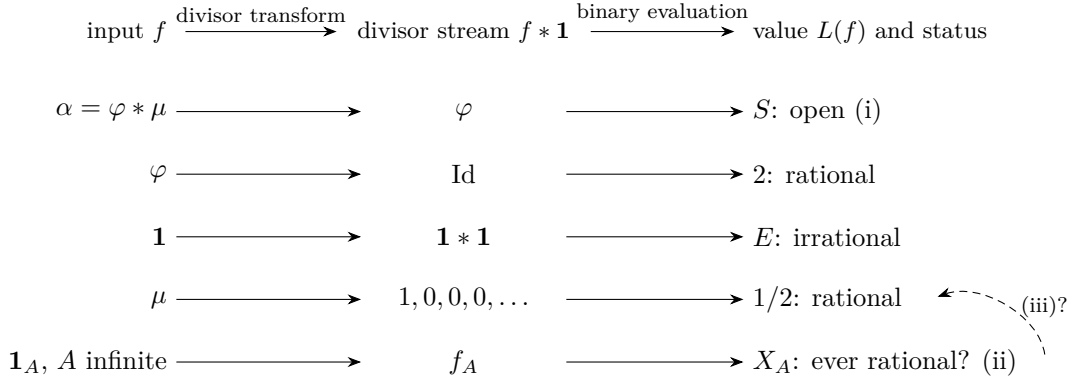


Figure 1: The one transform behind the paper, with the statuses of Section 3. Labels (i)–(iii) are the three questions of Section 6.1: #249 fixes the top input and asks about the single image point  $S$ ; universal #257 restricts the domain to indicators of infinite supports and asks whether its image ever meets  $\mathbb{Q}$ ; the half-value route fixes the value  $1/2$ , reached by the signed Möbius input through the stream  $1, 0, 0, \dots$ , and asks whether an infinite Boolean input reaches it as well. The figure organises statements proved in Sections 3–4 and Appendix B; it asserts nothing further.

## 6.2 Why no status transfers

The transform alone does not determine arithmetic status. The values formalised or cited in Section 3 include rational examples  $L(\varphi) = 2$  and  $L(\mu) = 1/2$ , the irrational example  $L(\mathbf{1}) = E$ , the transcendental example  $L(\text{Id})$ , and the open value  $L(\alpha) = S$ .

Simple properties of the input do not distinguish these cases. Both  $\varphi$  and  $\mathbf{1}$  are positive and multiplicative with at most linear growth, yet their transforms are respectively rational and irrational. Boolean coefficients alone are also insufficient: every finite support has rational value, while the infinite supports of Theorem 4.2 have irrational values. The finite example  $\{2, 3\}$ , whose value is  $10/21$ , and its period-six carry orbit are worked out in Appendix B.2.

A proof of an open case must therefore use arithmetic specific to its input: the dyadic structure of exact totient values for #249, the interaction between the Boolean Möbius condition and infinitude for universal #257, or the divisor feedback of the greedy orbit for the half-value fibre. Appendix C.3 rules out several arguments based only on generic input properties and identifies arithmetic information that a successful method would still need.

### 6.3 The shared bottleneck

Both reductions require finite events beyond every proposed cutoff, but the conclusions point in opposite directions. In literal quantifiers,

$$\begin{aligned} S \notin \mathbb{Q} &\iff \forall h \geq 1 \forall N_0 \exists N \geq N_0 \exists L \text{ Sep}(h, N, L), \\ \frac{1}{2} \in \mathcal{A} &\iff \forall N_0 \exists s \geq \max\{N_0, 6\} (\tau_s = 0). \end{aligned}$$

For  $S$ , Proposition 5.4 supplies fixed completeness, Theorem 5.5 gives the pointwise global form, and Theorem 5.6 gives the first displayed equivalence. For  $1/2$ , Theorem 4.11 gives the second: a zero terminal bit occurs beyond every row cutoff. In both cases the missing step is a theorem producing the required finite events beyond every cutoff. Neither such theorem is known. The classical full-support comparison is already proved: its Bertrand–CRT and averaging argument constructs the required witnesses uniformly at every precision. The coordinate identities and rational-support filters used elsewhere in the paper do not alter either missing quantifier.

Thus the comparison also gives a criterion for further progress. On the #249 side, a new local identity affects the open problem only if it produces or rules out **Sep**-certificates at unbounded parameters. On the half-value side of #257, it must produce terminal zeros beyond every row cutoff or prove that the greedy seam is eventually right. Further fixed-scale computation, by itself, changes neither open statement.

**What finite witnesses can decide.** For the half-value route, a finite calculation can settle nonmembership. By Theorem 4.14,  $1/2 \notin \mathcal{A}$  is equivalent to the existence of a fatal half-gap. Exact rational evaluation of the greedy prefix, together with the rational tail enclosures of Proposition B.13, can verify such a gap after finitely many steps. The gap excludes every support agreeing with that prefix. The reverse direction has no finite certificate here: survival through any finite row is inconclusive, while membership requires  $\tau_s = 0$  beyond every row cutoff.

For  $S$ , each **Sep**-certificate has narrower scope: it proves that one specified tail difference is non-integral. Theorem 5.6 requires certificates after every cutoff for every positive shift, so no single certificate proves irrationality. Rationality forces eventual integrality, but failure to find a certificate at one truncation proves neither integrality nor rationality: a greater binary depth  $L$  may still certify the same tail difference. Thus finite discovery can certify half-value nonmembership, but not half-value membership or either status of  $S$  within these certificate systems.

*Remark* (boundaries visible from the normal forms). The following two elementary counter-models are not formalised in the source. Both alter the object beyond a finite inspected range; neither makes a claim about the actual totient stream or the actual Mersenne weights.

For #249, any fixed finite family of certificate computations inspects only finitely many coefficients. The splice in Section 5.4 constructs a stream  $\gamma$  agreeing with  $\varphi$  on all of them, still satisfying  $0 \leq \gamma(n) \leq n$ , whose binary series is rational. The same finite output is therefore compatible with rationality. A proof of the missing supply must use totient values beyond every fixed bound.

For the half-value route, fix any finite prefix of greedy decisions and replace only the uninspected weights by a ratio-one-third geometric tail. If its total mass is below the pending residual, the complete modified tail falls short and the last inspected skip is final. If its total mass equals the residual, that tail realises the target. Both choices preserve strict tails and every inspected decision. Thus a finite prefix and strict-tail geometry do not decide membership; a proof must use the specific Mersenne arithmetic of the tail.

## 7 Formal verification and reproducibility

This section records proof authority and reproducibility; it does not add a mathematical conclusion to either reduction.

**Proof authority and source identity.** The Lean 4 proof assistant [21] and Mathlib library [22] check the named results. The source, exact informal-to-formal links, and build instructions are available in the [public repository](#). This paper pins formal-source revision `ebc98ee71f68`, published under the tag `formal-source-2026-07-18`. The blue links resolve to immutable source files at that checkpoint. The release also pins the Lean and Mathlib versions; the toolchain is Lean `v4.29.1`.

**Reproduction commands and artefact identities.** A cold clone at the published revision is checked with `lake build`. The release-validation command also checks that every registered claim resolves to its stated Lean declaration and manuscript location, that the PDF matches the committed source, and that the public source pin resolves to the recorded revision. The resulting report records release validation; it is not mathematical proof authority. The repository records SHA-256 identities for the claims file, Lean-declaration index, manuscript source, rendered PDF, and generated navigation files, together with their regeneration commands. No hardware-independent build time is claimed; elapsed time depends on the Lean cache and machine, and is not part of the mathematical claim.

**What the computation establishes.** Lean elaborates each declaration into a proof term, which the kernel checks; tactics construct these terms but are not separate proof authority. `norm_num` constructs concrete arithmetic proofs, and `decide` reduces a decidable finite proposition to obtain a proof. The [finite survivor-count theorem](#) uses `decide` to count 412 phases. Its local `maxRecDepth` setting only raises the permitted unfolding depth; the proposition and logical rules are unchanged. It does not use compiler-backed native evaluation, and its adjoining `#print axioms` audit reports neither `Lean.trustCompiler` nor `sorryAx`, Lean’s unfinished-proof axiom. Generated Lean certificate declarations receive the same check. An external dataset, Python release script, generated index, or this manuscript does not. Appendix E connects each displayed result to its exact formal statement.

**Section consequence.** At the published formal-source checkpoint, the linked declarations and their checked proof terms certify the stated proved, conditional, finite, and equivalence results. This verification does not change the mathematical status of the unresolved statements: the open problems collected in Section 8 remain open.

## 8 Exact remaining problems and further questions

Section 2 gives the exact reductions. Here we record two concrete sufficient criteria and then state the unresolved questions. Erdős #249 has one open obligation in two equivalent forms. Erdős #257 has a universal statement and a counterexample route pointing in opposite directions: half-value membership would refute the universal statement, while nonmembership would decide only the half-value question.

### 8.1 Two concrete scalar criteria

Two scalar conditions give concrete sufficient routes toward the two open directions. Both are exact arithmetic statements; neither is proved at the unbounded range needed for its final conclusion.

For #249, retain  $M_t = \text{lcm}(1, \dots, t)$  and write

$$P_L(N) = \sum_{j=0}^{L-1} \varphi(N+1+j) 2^{L-1-j}.$$

Thus  $P_L(N)$  is the base-two weighted block of the next  $L$  totient values after  $N$ . At scale  $t$ , choose one logarithmic depth and compare the blocks after  $M_t$  and  $2M_t$ :

$$m_t = \lfloor \log_2 M_t \rfloor + 10, \quad d_t = (P_{m_t}(2M_t) - P_{m_t}(M_t)) \bmod 2^{m_t},$$

where  $0 \leq d_t < 2^{m_t}$ . The integer  $d_t$  is their block-difference residue. Define

$$\sigma_t = \min(d_t - 2^{m_t-5}, (2^{m_t} - 2^{m_t-5}) - d_t) \in \mathbb{Z}.$$

This is the signed distance from the nearer edge of the central band below: it is nonnegative exactly on that band and negative outside it.

**Proposition 8.1** (fixed-depth central-band equivalence). *For every  $t \geq 1$ , the residue  $d_t$  lies in the central band  $[2^{m_t-5}, 2^{m_t} - 2^{m_t-5}]$  if and only if  $\sigma_t \geq 0$ .*

This fixed-scale equivalence and its beyond-every-cutoff strict-jump form are the [canonical central-band equivalence](#) and [strict-jump slack equivalence](#).

**Corollary 8.2** (strict-jump slack criterion). *If, for every  $t_0$ , there is a  $t \geq \max\{3, t_0\}$  such that*

$$M_t < M_{t+1} \quad \text{and} \quad \sigma_t \geq 0,$$

*then  $S$  is irrational.*

This implication is the [strict-jump slack irrationality criterion](#). The proposition is a fixed- $t$  equivalence; the corollary gives only sufficiency. No converse is known: completeness may choose a depth from the tail difference, whereas this criterion fixes  $m_t$ . Persistent negative slack at that depth refutes only this route and does not decide whether  $S$  is rational.

For #257, write  $r_n = r_n(1/2)$  for the unrepresented remainder after rank  $n$ . The exact tail expansion and second-scale normalisation are

$$T_n = 2^{-n} + \frac{1}{3} 4^{-n} + E_n, \quad u_n = 4^n(2r_n - 2^{-n}),$$

where  $0 \leq 4^n E_n \leq (2/7)2^{-n}$ . The identity  $u_n - 1/3 = 4^n(2r_n - T_n) + 4^n E_n$  explains the centre  $1/3$ : distance from it is the scaled residual-tail imbalance, up to the bounded error. Under  $r_n \leq T_n$ , the next step survives exactly when  $|2r_n - T_n| \geq w_{n+1} - T_{n+1}$ . A sufficient condition is

$$\frac{1}{6} + \frac{37}{56} 2^{-n} \leq \left| u_n - \frac{1}{3} \right|.$$

We call this the *second-channel separation condition*. Here  $37/56 = 2/7 + 3/8$  budgets the higher-tail and gap errors, respectively. The absolute value is needed because  $u_n < 0$  at  $n = 3, 4$ .

**Proposition 8.3** (finite second-channel separation). *The second-channel separation condition is decidable over  $\mathbb{Q}$  and holds for  $1 \leq n \leq 6$ .*

This is the [finite second-channel separation](#).

**Corollary 8.4** (second-channel separation criterion). *If the second-channel separation condition holds for every  $n \geq 7$ , then  $1/2$  belongs to the Mersenne achievement set  $\mathcal{A}$ .*

This implication is the [conditional from-seven membership criterion](#). The finite proposition supplies  $1 \leq n \leq 6$ , but no case with  $n \geq 7$ ; no converse is proved. Membership itself has the exact characterisation

$$1/2 \in \mathcal{A} \iff \text{the canonical greedy orbit omits infinitely many exponents.}$$

This is formalised at [half-membership-infinite-skip equivalence](#). The forward implication uses the full-support irrationality theorem; for the reverse, if a residual ever exceeds the remaining tail, every later exponent is selected; hence only finitely many omissions remain. The  $n \geq 7$  hypothesis is open. If proved, it would yield an infinite support of rational value  $1/2$ , refuting rather than proving the universal #257 statement.

**Dependency direction.** The two criteria have the proved logical form

|                                                                                                                                                                    |                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| after every cutoff, nonnegative slack<br>occurs at a strict lcm jump<br>$\Downarrow$<br>unbounded #249 certificate supply<br>$\Downarrow$<br>$S \notin \mathbb{Q}$ | second-channel separation<br>for every $n \geq 7$<br>$\Downarrow$<br>$\frac{1}{2} \in \mathcal{A}$<br>$\Updownarrow$<br>$S(1/2)$ is infinite. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

Neither premise is proved. The range  $1 \leq n \leq 6$  is only a finite prefix of the second. The left conclusion settles #249; the right would refute, not prove, the universal #257 statement.

## 8.2 The unresolved statements

**Erdős #249: one obligation in two equivalent forms.**

**Problem 8.5** (Erdős #249). Prove that

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

is irrational.

By Theorem 5.6, the preceding problem is equivalent to the following exact quantifier statement.

**Problem 8.6** (unbounded certificate supply). Prove (5.1). Equivalently, prove the lcm-diagonal supply in Theorem 5.7. The finite scales in Example 5.8 do not establish either unbounded form.

These are not two independent open problems: Theorem 5.6 and Theorem 5.7 identify the second as an exact certificate form of the first.

**Erdős #257: universal statement and counterexample route.**

**Problem 8.7** (universal Erdős #257). Prove that

$$\sum_{n \in A} \frac{1}{2^n - 1}$$

is irrational for every infinite set  $A \subseteq \mathbb{N}_{>0}$ , rather than only for the support families in Theorem 4.2.

**Problem 8.8** (the half-value counterexample route). Decide whether  $1/2 \in \mathcal{A}$ . By Theorem 4.11, this is equivalent to the canonical greedy orbit omitting infinitely many exponents. A positive answer would give an infinite support of rational value  $1/2$  and refute the universal #257 statement; a negative answer would decide only this distinguished point-membership question.

**What the reductions change.** The reductions change form, not status. For #249 the one missing theorem is unbounded certificate supply, equivalently in pointwise or lcm-diagonal form. For the half-value route to #257, arbitrary supports give way to one greedy orbit; in final-skip coordinates it remains to exclude cells  $-2, -1$  and prove the future-tail inequality at the remaining nonnegative-carry middle transitions. Supply beyond every cutoff proves #249; terminal zero beyond every row cutoff refutes universal #257; eventual right behaviour proves only  $1/2 \notin \mathcal{A}$ .

**Two refinements not pursued.** Certificate completeness gives a least successful depth at every non-integral pair, but no growth bound in  $(h, N)$ ; such a bound would decide each fixed pair. Separately, the zero-window theorem is inert when a support contains a small element (Appendix B.3); coverage outside every finite core would remove this. We have not pursued either question.

## Statements and declarations

**Artefact and data availability.** The [archived formal-source revision](#) contains the Lean sources, pinned toolchain, library manifest, and generated certificate data used in the verification. Repository metadata and this manuscript provide navigation rather than proof authority; Section 7 gives the verification route.

**Use of AI assistance.** Large-language-model assistants were used during development for proof drafting, refactoring, and prose editing. Every registered formal claim is linked to a declaration in the archived source revision. Lean checks each proof term against the pinned library; the project contains no proof placeholders or project-defined axioms.

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**Acknowledgements.** The problem numbering and status follow the Erdős Problems catalogue maintained by Thomas Bloom [23].

## A Technical reductions for the half-value branch

This appendix records the local arithmetic used to reach Theorems 4.11 and 4.14, Proposition 4.15, and Corollary 4.16. Its statements have four statuses: exact identities or equivalences; implications with an unproved hypothesis; exact finite calculations; and counterexamples showing that a proposed local argument is insufficient. None changes the two open conclusions at the end of Section 4. Readers concerned only with the main reductions may skip the appendix.

The parts are ordered by their role in the greedy argument. Appendix A.1 computes the residual interval for one take-skip-take pattern, without showing that the half-value orbit enters or avoids it. Appendix A.2 gives one conditional irrationality criterion for a structured support, several sufficient hypotheses for  $1/2 \in \mathcal{A}$ , and the local identities they use. Appendix A.3 separates a finite witness from the unbounded supply of such witnesses needed for membership. Appendix A.4 identifies nonmembership with a last greedy skip, excludes some possible last transitions, and states the two cell exclusions and tail inequality still missing. Appendix A.5 contains finite checks, changes of coordinates, and counterexamples to shortcuts.

The two routes used below have the logical form

$$\begin{aligned} \text{A.2--A.3 unbounded-depth witnesses} &\implies \frac{1}{2} \in \mathcal{A} && \implies \text{infinite support summing to } \frac{1}{2}, \\ \text{A.4 exclusions and tail bound} &\implies \text{no final skip} && \iff \frac{1}{2} \in \mathcal{A}. \end{aligned}$$

The displayed implications are proved, and the final equivalence is exact. Both leftmost hypotheses are open. Thus the appendix explains what would suffice and why several finite arguments fail; it does not decide whether  $1/2 \in \mathcal{A}$ .

Throughout this appendix:

- a *take* (respectively a *skip*) at rank  $n$  is the greedy decision  $\varepsilon_n = 1$  (respectively  $\varepsilon_n = 0$ );
- the *finite greedy seam at row  $s$*  is the Boolean word of Definition 4.10; an unqualified *seam* in this appendix is an adjacent pair of such words at a branch boundary, and the compounds seam pair, one-hole seam, first-wrap seam, and rewind seam carry this pair sense;
- the *square-root strip* at depth  $N$  is the carry interval  $[1, H_N]$  with  $H_N = 2\lfloor\sqrt{N}\rfloor + 4$ ; an interval of carries kept available by a construction is *protected*;
- a *hole* of a seam is the missing carry value in an otherwise covered protected interval;
- a *rewind* is the inverse finite recurrence of a seam;
- the *transition carry* is the integer coordinate appearing in the exact residual identity, and the *remainder* of a row is its integer greedy residual;
- the *pulse* of a rank at row  $s$  counts its divisor incidences at  $2s + 1$  and  $2s + 2$ , the first counting twice; the pulse of a word is the sum over its ranks.

These terms denote mathematical objects, not stages of a proof search. Module-local coordinates not listed here (for example the centred, lazy, or first-wrap variants) are defined in the linked formal statements.

## A.1 Exact band for a final skip between two takes

At rank  $k$ , call the current residual *dyadically safe* when it is at most  $2^{-k}$ , and *unsafe* otherwise. Since the actual Mersenne tail satisfies  $T_k > 2^{-k}$ , safety is sufficient but not necessary for survival; an unsafe skip need not stop the orbit.

Suppose a take at rank  $b$  is followed by the next take at  $c > b + 1$ . Put  $q = 2^b - 1$ ,  $m = 2^{c-1}$ , and write the residual just before rank  $b$  as  $\rho = 1/R$ , so  $R$  is its reciprocal. Taking  $w_b = 1/q$  changes the reciprocal to  $R' = Rq/(q - R)$ , which stays fixed through the intervening skips. The last skip, at rank  $c - 1$ , occurs exactly when  $R' > m - 1$  and is dyadically unsafe exactly when  $R' < m$ . For  $0 < R < q$ , solving  $m - 1 < R' < m$  for the pre-take reciprocal gives

$$\frac{q(m-1)}{q+m-1} < R < \frac{qm}{q+m}.$$

The interval has exact width

$$\frac{q^2}{(q+m)(q+m-1)}.$$

Formalised: [the general unsafe-band equivalence](#) and [the general band-width formula](#).

For a single intervening skip,  $c = b + 2$  and  $m = 2q + 2$ . The band becomes

$$\frac{q(2q+1)}{3q+1} < R < \frac{2q(q+1)}{3q+2},$$

with width  $q^2/((3q+1)(3q+2)) < 1/9$ , entirely between  $2q/3$  and  $2q/3 + 2/9$ . No integral  $R$  lies in this window. If instead  $\rho = p/(2D)$ , then  $R = 2D/p$ ; for positive odd  $p, D, q$ , clearing denominators in the same inequalities forces  $p \geq 7$ . Formalised: [the one-skip band equivalence](#), [the one-skip width formula](#), [the width bound](#), [the integral-reciprocal exclusion](#), and [the odd-coordinate lower bound](#).

These results neither show that the greedy orbit of  $1/2$  enters or avoids the band nor identify the remaining transition coordinates  $-2, -1$  with dyadically unsafe skips.



## A.2 Conditional criteria and local inputs for structured supports

The results here have three distinct roles. The bouquet criterion is a conditional irrationality theorem for one structured support family. The seam, strip, and terminal criteria are sufficient hypotheses for  $1/2 \in \mathcal{A}$ . The dilation, prime-power, and return-carry results are local algebraic inputs that prove neither an irrationality conclusion nor half-membership without an additional supply hypothesis. Nothing in this subsection is a new unconditional case.

### A.2.1 A conditional pairwise-coprime dilation criterion

Write  $A = E \cup \{c_i p_i : i \geq 0\}$ , where  $E$  is a finite set of positive integers and  $Q, c_i, p_i$  are positive. Suppose that  $Q$  is divisible by every  $d \in E$  and every  $c_i$ , while  $p_i > 1$  is coprime to  $Q$  and to  $p_j$  for  $j \neq i$ , and  $\sum_i 1/p_i < \infty$ . We call this a *finite-core orthogonal-petal bouquet*:  $Q$  carries the fixed factors and the  $p_i$  are mutually coprime. Lean includes the example  $E = \emptyset$ ,  $Q = 6$ , with  $c_i$  alternating between 2, 3 and  $p_i = 5^2, 7^2, 11^2, \dots$ ; each support element has three prime factors with multiplicity.

Put  $f_A(n) = \#\{d \in A : d \mid n\}$  and  $T_A(M) = \sum_{j \geq 1} f_A(M+j)2^{-j}$ . For every  $K \geq 1$ , the unproved selector requires some  $N$  for which  $2^K \mid \sum_{r=1}^K f_A(N+r)2^{K-r}$  and  $T_A(N+K) \leq 16$ . These are the carry divisibility and scaled-tail bounds. Lean proves that they imply the [forced-carry certificate supply](#) and hence [irrationality of the support series](#). The selector itself is not proved. This is therefore a conditional reduction for one structured support family, not an unconditional case of Erdős #257.

### A.2.2 Composite dilation defects

For the same divisor count  $f_A$ , multiplication by a composite support element  $a$  has an exact correction term. Besides the distinguished divisor  $a$ , the new divisors are counted by a finite defect which excludes divisors already present at  $x$ . The formal identity is [the composite-dilation identity](#). For a finite-core orthogonal-petal bouquet, the defect at a ray is bounded by the finite exceptional frame plus the divisor count of the petal support: [the bouquet defect bound](#). This records the local correction needed when dilation is not prime. It does not prove a correlation estimate, the sunflower tail selector, or an irrationality statement.

### A.2.3 Mixed prime-power layers

The one-prime-power layer has a two-signature extension. Exact  $p$ -adic and  $q$ -adic layer operators commute, and, for distinct primes with the residual argument coprime to  $pq$ , their composite extracts the iterated exact-valuation pullback of the support coefficient. This is formalised by [the two-prime layer identity](#). The singleton support  $\{12\}$  gives a checked  $(2^2, 3)$  example with value one at residual argument 1: [the singleton-support calculation](#). These are finite coefficient identities. They do not provide a decimation transport, a bounded- $\Omega$  conclusion, or an irrationality theorem.

### A.2.4 Half-trapping return carries

If rational-linear channels vanish on every relation killed by one scalar evaluation, their evaluation matrix has rank at most one; every nontrivial square minor therefore vanishes. The all-rank determinant statement is [the rank-one determinant criterion](#). For two reverse carries with a common coefficient word, a 1/0 seam, and a common following overlap, the terminal difference is a power of two times an odd integer. A shared nonnegative Archimedean bound then gives the matching dyadic spacing inequality, formalised at [the dyadic spacing inequality](#). The argument does not produce reverse-carry presentations or an earlier overlapping return; it supplies only the algebraic core of that conditional criterion.

### A.2.5 Half-carry reachability

There is also a finite-prefix criterion for the target  $1/2$ . At an unresolved even seam, the scalar transition reaches every integer terminal carry except the single value  $2\delta - c$ ; this is the exact equivalence [the even-seam reachability equivalence](#). A canonical description of such a seam at every sufficiently large even depth is an unproved hypothesis. Under that hypothesis, the finite admissible words are cofinal, by [the cofinal-admissibility implication](#), and compactness then produces an infinite support with sum exactly  $1/2$ : [the half-value compactness conclusion](#). Thus the formal result is a conditional reduction. It does not establish the canonical seam supply and does not prove Erdős #257.

**Cofinal strip returns.** The all-level strip hypothesis can be weakened for the canonical greedy half carry. Its uncentred, binary-scaled form is antitone. Consequently, if it returns to the square-root strip cofinally, then one sufficiently late return bounds every later scaled carry; the strip envelope divided by  $2^N$  tends to zero. The resulting tempered centred carry yields a support selected by the greedy expansion whose series value is  $1/2$ ; the finite-support exclusion makes that support infinite.

**Proposition A.1** (cofinal strip-return criterion). *Let  $G_M$  be the uncentred integer carry of the canonical greedy expansion of  $1/2$ . If*

$$\forall N \in \mathbb{N} \exists M \geq N, \quad G_M \leq 2\lfloor \sqrt{M+1} \rfloor + 4,$$

*then  $1/2 \in \mathcal{A}$ , and its canonical greedy support is infinite.*

Formalised: [scaled-carry decay](#) and [the strip-return membership criterion](#). The cofinal return condition itself is not established. It is weaker than a uniform strip bound and still does not give an unconditional membership theorem.

**Terminal-only cofinal approximation.** Coherence can be removed entirely from a further version of this reduction. At depth  $M$ , take only a normalised finite word whose terminal half carry lies in the square-root strip. Its finite support then has support-series value within

$$\frac{4M + 12}{2^M}$$

of  $1/2$ . If such terminal witnesses occur cofinally, these finite values converge to  $1/2$ ; closedness of the Mersenne achievement set supplies the limit. No control of the earlier rows, and no compatibility between the witnesses, is used.

**Proposition A.2** (terminal-only cofinal criterion). *Cofinal normalised terminal witnesses in the square-root strip imply that  $1/2$  belongs to the Mersenne achievement set, and hence that an infinite support has support-series value  $1/2$ .*

Formalised: [the terminal approximation bound](#), [the terminal-strip compactness criterion](#), and [the infinite-support conclusion](#). The cofinal terminal-witness supply is not proved. In particular, the proposition does not establish membership of  $1/2$  unconditionally.

**Scaled terminal vanishing.** The square-root strip is stronger than the analytic condition used by the terminal-only argument. Let the depths of normalised finite words tend to infinity. If their terminal carries, after division by the corresponding power of 2, tend to zero, then the universal coefficient-tail error also tends to zero. The finite support values therefore converge to  $1/2$ , and closedness of the achievement set again supplies the limit. The earlier terminal-only strip condition implies this weaker sequential hypothesis.

**Proposition A.3** (scaled terminal-vanishing criterion). *A normalised terminal-word sequence with depths tending to infinity and vanishing binary-scaled terminal carries yields an infinite support with support-series value  $1/2$ .*

Formalised: [the scaled terminal error bound](#), [the scaled-vanishing compactness criterion](#), and [the scaled-vanishing support conclusion](#). No such scaled-vanishing sequence is constructed here; the result is a conditional reduction, not an unconditional membership theorem.

### A.3 Finite-to-cofinal criteria: windows, seams, and compactness

Every half-value route in this subsection factors as

$$\text{finite witness at depth } N \quad + \quad \text{a cofinal supply of such depths} \quad \implies \quad \frac{1}{2} \in \mathcal{A}.$$

The local identities and recurrences describe or advance one finite witness. The compactness statements turn a cofinal supply into membership. No result below proves that supply, so a verified window or seam at one depth is evidence for a possible recurrence, not an unbounded theorem.

#### A.3.1 Finite residuals for the half-value problem

For an admissible finite word, the residual from  $1/2$  is given exactly by its terminal half carry minus the future coefficient tail, scaled by a power of two; see [the finite-shadow residual identity](#). At a first seam, avoiding the integer strip is equivalent to full scalar reachability of that strip: [the seam-escape equivalence](#). The rank-three greedy example is raw dyadically safe while its seam has an in-strip hole, so raw safety alone does not imply this escape. Finally, an explicit frozen-margin hypothesis implies  $1/2$  belongs to the Mersenne achievement set, by [frozen-margin implication](#). That hypothesis is unproved; this is a conditional reduction, not a proof of Erdős #257.

#### A.3.2 Fixed-coefficient rewind

For a finite list of common coefficients, the canonical inverse-parent map has an exact closed form with a power-of-two denominator: [the rewind closed form](#). An interval of width at most that denominator contracts either to one ancestor or to two adjacent ancestors. The singleton alternative is equivalent to the explicit dyadic phase inequality, while the other alternative is a seam pair; see [the singleton-or-seam dichotomy](#). Thus width alone does not justify a singleton conclusion. This is a finite rewind lemma, and supplies neither a seam exclusion nor an Erdős #257 conclusion.

#### A.3.3 Selected half-carry windows

The selected-window construction keeps an explicit admissible word for each terminal carry in a finite protected interval. A supply of such windows at all sufficiently large levels gives cofinal admissibility and hence an infinite support with sum  $1/2$ : [selected-window cofinality](#) and [the selected-window half-value conclusion](#). The unresolved input is a construction of such a window at every sufficiently large depth. The later propagation results require either common next-row divisor data or an adjacent seam. None constructs the required window supply, so these theorems do not prove Erdős #257.

#### A.3.4 A realised rewind criterion

For a realised fixed-coefficient history, the dyadic rewind phase is the complementary binary suffix numeral: [the suffix-phase identity](#). Consequently the singleton criterion is exactly the

inequality that this suffix numeral covers the rest of the terminal interval, formalised by [the suffix-coverage criterion](#). This converts the phase condition into support-history coordinates, but does not prove the required coverage inequality globally.

### A.3.5 Localised protected seams

The compactness argument needs only one admissible word at cofinally many depths. At a localised one-hole seam, one of the two protected carries 3 and 4 survives; therefore a cofinal supply of realised protected seams implies cofinal admissibility by [the protected-seam cofinality implication](#), and hence an infinite support with sum  $1/2$ , by [compactness](#). The cofinal realisation supply is not proved.

### A.3.6 Combining the two cofinal half-carry criteria

Assume that at each requested scale there is either a selected protected window or a localised realised seam. Each alternative supplies the one admissible word needed for compactness, so this cofinal disjunction implies an infinite support with sum  $1/2$ : by [window-or-seam cofinality](#) and by [the window-or-seam half-value conclusion](#). The cofinal disjunction remains unproved.

### A.3.7 Integer-greedy first-wrap reduction

The first-wrap seam has exact truncated Mersenne weights with a quantitative gap-dominance property. For these weights, a small positive subset-sum defect exists exactly when the descending integer-greedy remainder is small and positive; this is formalised by [the integer-greedy remainder equivalence](#). The remaining arithmetic input is an unbounded lower bound for that deterministic remainder sequence. No such lower bound is proved here.

### A.3.8 Exact finite recurrence for the greedy seam

The finite seam-word construction realises the abstract adjacent cut by concrete Boolean words and identifies its next word with the integer-greedy choice: [the seam-word recurrence](#). It implements a finite recurrence, not the unbounded remainder estimate.

## A.4 Exact final-skip reduction and its residual cells

This is the only subsection that feeds the main final-skip reduction directly. It first identifies nonmembership with a final skip and membership with cofinal terminal zeros. It then separates the terminal cells already excluded from the two unresolved middle coordinates and the unresolved future-tail inequality. The later largest-skip, pulse, and reset formulas are conditional ways to attack those residual obligations; they do not add a second conclusion.

### A.4.1 Fatal gaps and eventual right tails

For the concrete integer seam words, eventual right extension is equivalent to nonmembership of  $1/2$  in the Mersenne achievement set: [the eventual-right nonmembership equivalence](#). The same condition is equivalent to the existence of a finite fatal half gap, by [the fatal-gap equivalence](#), and also to a final skipped exponent in the real greedy orbit, by [the final-skip equivalence](#). Thus this equivalence gives an exact classification of the nonmembership branch. It neither decides whether  $1/2$  belongs to the achievement set nor proves the universal Erdős #257 statement.

This is the nonmembership half of the argument summarised earlier: a last greedy skip is equivalent to eventual right extension. The following paragraph records the first exceptional middle case that can be ruled out.

**First final-middle cell.** The first exceptional middle cell has a separate obstruction: if the orbit extends right after the last skipped rank, then the cell with transition carry  $-3$  makes the lazy cofinite tail strictly smaller than  $1/2$  while its centred carry becomes negative. This is impossible by the analytic nonnegativity identity, [the middle- \$-3\$  exclusion](#). It removes that one conditional terminal configuration; it does not decide the right-tail branch or the remaining two middle cells.

**Second final-middle cell.** The next cell, with transition carry  $-2$ , is not eliminated. Here its *phase* means the triple of residue classes of the putative last skip  $D$  modulo 21, 10, and 13. The possible phases are nevertheless sharply constrained. Uniformly for seam rows  $s \geq 27$ , the greedy decisions through rank 26 select exactly the ranks

$$2, 3, 6, 7, 14, 20, 21, 26,$$

by [the exact rank-\(26\) prefix](#).

If  $D \geq 27$  lies in this middle cell and is followed by an all-right suffix, then

$$D \bmod 21 \in \{11, 14, 17, 20\}, \quad D \bmod 10 \neq 8, \quad D \bmod 13 \neq 11,$$

the class  $D \bmod 10 = 7$  cannot pair with  $D \bmod 21 = 11$ , and the class  $D \bmod 13 = 10$  cannot pair with  $D \bmod 21 = 11$ . The orbit theorem is [the final-middle phase sieve](#). The three moduli are pairwise coprime, so their phase triples correspond to the  $21 \cdot 10 \cdot 13 = 2730$  residue classes modulo 2730. Exactly 412 survive; this finite count is [the exact survivor count](#). This is a uniform exact reduction of the  $-2$  cell, not its exclusion; the surviving phases, the  $-1$  cell, and the right-tail branch remain open. A smaller survivor proportion at deeper ranks would still not exclude the cell. A nested system of nonempty residue classes can have vanishing density and a nonempty inverse limit. This route must therefore either reach finite extinction at some rank or add an Archimedean estimate at a surviving phase.

The remaining global reduction is now precise. If every genuine middle transition other than the already discharged  $C_D = -3$  cell has transition carry larger than its complete future divisor-incidence tail, then an eventual right seam is impossible and  $1/2$  belongs to the Mersenne achievement set: [middle-tail implication](#). The stronger square-root lower-bound hypothesis is also sufficient, [square-root sufficient condition](#). Neither hypothesis is supplied here.

For finite seam prefixes the remaining tail bound sharpens further: the future divisor-incidence tail is bounded by the cardinality of the prefix. Thus the explicit row-scale inequality  $\#G_D + \text{belowPulse}_D + 5 < 4 \text{remainder}_D$  is sufficient for the same half-membership conclusion. The hypothesis is [the cardinality escape condition](#), and the resulting membership theorem is [the prefix-cardinality bound](#); it is enough in turn that every late middle remainder is at least its row, [row-scale sufficient condition](#). These are exact conditional reductions, not proofs of the row bound.

**Finite upper-reset band certificates.** The companion upper branch has a kernel-checked finite certificate: for every upper transition from rows 13 through 30, every subsequent dyadic danger band is avoided. The certificate shares each exact successor remainder across all its band indices, [the verified upper-reset certificate](#). It is a bounded transition check and does not supply the cofinal band-escape hypothesis.

#### A.4.2 Positive half-membership classification

The complementary branch has an equally exact seam formulation. Membership of  $1/2$  in the Mersenne achievement set is equivalent to the occurrence, beyond every bound, of a successor terminal bit equal to zero, [the unbounded terminal-zero equivalence](#); equivalently, it is equivalent to a cofinal sequence of such terminal zeros, [the cofinal terminal-zero equivalence](#). It is

also equivalent to a cofinal integer-seam sequence with skipped ranks tending to infinity, [the unbounded skipped-rank equivalence](#). These equivalences isolate the required cofinal skipped-rank condition; they do not establish that condition.

#### A.4.3 Largest skipped ranks

At a named largest skipped rank  $d$ , the lower seam word and its adjacent upper competitor have an exact late-range gap. In division-free form, the identity is

$$3L + (3 \cdot 2^{s+1} + 2 \cdot 4^{s-d} + 4) = 3U$$

whenever  $2s < 3d$ ; this is formalised at [the exact late-gap identity](#). The upper and middle successor branches create a terminal largest skipped rank, while the right branch preserves an existing one: [the terminal largest-skip branch](#) and [right-branch persistence](#). The first-crossing branch hypothesis required to use this identity globally is not proved.

#### A.4.4 Boundary-pulse normalisation

Above a late largest skipped rank, every filled suffix rank has zero row pulse, so the word pulse is carried entirely by the lower prefix: [the lower-prefix pulse localisation](#). At the first crossing of the two-thirds boundary there are exactly two cases,  $3d = 2s + 1$  and  $3d = 2s + 2$ , and the skipped rank has pulse respectively two or one: [the first-crossing pulse classification](#). This supplies the incidence normalisation for a transition-carry estimate; it does not establish that estimate.

#### A.4.5 Largest-skip induction

Assume that at the first crossing a late largest skipped rank either remains late at the next row or forces an upper-or-middle successor branch. This hypothesis propagates a late largest skipped rank from row 14 onward, [largest-skip propagation step](#), and hence yields cofinal skipped ranks and membership of  $1/2$  in the Mersenne achievement set, [largest-skip membership criterion](#). The first-crossing hypothesis itself remains unproved.

#### A.4.6 The fixed-tail survival criterion

A skipped rank is final exactly when its post-decision remainder lies strictly above the complete remaining Mersenne tail: [the final-skip criterion](#). Consequently membership of  $1/2$  is equivalent to the pointwise assertion that every actual skipped rank has remainder at most its full future tail, [the half-membership survival equivalence](#). This is an exact local reformulation; the tail-survival inequality is not proved.

#### A.4.7 The carry identity at a non-right transition

For a finite support aligned with a non-right transition, the residual from  $1/2$  is exactly the integer transition carry minus the complete future divisor-incidence tail, scaled by  $2^{-(2d+2)}$ : [transition-carry residual identity](#). Thus the carry exceeds that tail exactly when the support series lies below  $1/2$ , and a negative carry places it above  $1/2$ ; [positive-carry criterion](#) and [negative-carry criterion](#). An upper carry at most  $-8$  even gives a full Mersenne-gap margin, [full-gap carry criterion](#). These are local criteria, not a proof that the required carries occur.

#### A.4.8 Quarter-band endpoint cells

For the adjacent seam recurrence, failure of the desired next-row upper bound on the middle branch is exactly one multiple-of-four pulse cell, [the middle-branch pulse-cell equivalence](#). On



the terminal-reset branch, the same failure is exactly a constrained  $3 \cdot 2^{s-1} + k$  pulse cell, [the reset-branch pulse-cell equivalence](#). These are arithmetic normal forms for the exceptional cells; no avoidance statement for the concrete seam orbit is proved.

#### A.4.9 Reset-deficit escape

At a late largest skipped rank, a right successor branch pins the integer greedy remainder between an explicit quarter threshold and a carry threshold: [the right-branch remainder window](#). Along such a branch the remainder satisfies an exact affine recurrence, [the right-branch affine recurrence](#). Moreover, a doubled-scale deficit forces an upper-or-middle branch within the corresponding number of rows, [the deficit escape bound](#). The remaining input is the deterministic anti-concentration needed to exclude the pinned right-branch window globally; no such anti-concentration statement is proved here.

#### A.4.10 Alignment of seam and transition carries

At a first-wrap seam, the transition carry of the left boundary support is four times the shifted seam hole, less the explicit paired incidence pulse at the next two rows: [seam-carry alignment](#). This is an exact coordinate change between the seam and transition coordinates; it does not supply a sign or size estimate for that carry.

### A.5 Finite evidence, coordinate changes, and eliminated special cases

This subsection decides no open question. Its verified bases and stages are bounded evidence for the cofinal criteria in A.2–A.3; its coordinate changes translate one finite formulation into another; and its counterexamples and incompatibility theorems rule out shortcuts that would otherwise be mistaken for global arguments. Each item must therefore be read through the earlier criterion it supports or blocks.

#### A.5.1 A verified initial selected window

The selected half-carry construction has a checked depth-18 base: twelve Boolean words cover every terminal carry in the strip  $[1, 12]$ , with all intermediate carries in the corresponding half strip. The reflected table is formalised at [the depth-18 table verification](#); the resulting selected window is defined at [the depth-18 selected window](#). All twelve words share the same restriction at depth 13, giving the first next-row divisor-agreement certificate, [the depth-18 divisor-agreement certificate](#). This finite base does not provide selected windows at unbounded depths.

#### A.5.2 Rewind-to-seam equivalence

For a selected window with a rewind seam, a two-value coefficient profile on the adjacent base ancestors induces the live next-row profile, [the next-row rewind profile](#). Under the stated strip and buffer inequalities, that profile realises a protected one-hole seam at the next row, [the protected-seam realisation](#). The base-ancestor coefficient drop is an explicit input, not a theorem here, and so are the stated strip and buffer inequalities.

#### A.5.3 Selected suffix cylinders

Put

$$H_N = 2\lfloor\sqrt{N}\rfloor + 4.$$

For a Boolean word  $a = (a_0, \dots, a_N)$ , let

$$f_a(n) = \#\{d \leq N : d \mid n, a_d = 1\}, \quad K_0(a) = 1, \quad K_{r+1}(a) = 2K_r(a) - f_a(r+2).$$

**Definition A.4** (selected window and suffix cylinder). A *selected window* of depth  $N$  and radius  $R$  is a family  $(a^{(k)})_{1 \leq k \leq R}$  of Boolean words with  $a_0^{(k)} = a_1^{(k)} = 0$  such that

$$1 \leq K_{n-1}(a^{(k)}) \leq H_n \quad (1 \leq n \leq N), \quad K_{N-1}(a^{(k)}) = k.$$

For  $M \leq N$ , define the suffix numeral

$$\nu_{M,N}(a) = \sum_{j=M+1}^N a_j 2^{N-j}.$$

The window has a *suffix cylinder at cutoff  $M$  with endpoint  $E$*  if all  $a^{(k)}$  have the same prefix through  $M$  and

$$\nu_{M,N}(a^{(k)}) + k = E \quad (1 \leq k \leq R).$$

A *profiled gap stage* permits one missing carry interval  $[g_-, g_+] \subseteq [1, H_N]$ : lower and upper suffix-cylinder families represent  $1, \dots, g_- - 1$  and  $g_+ + 1, \dots, H_N$ , with independent fixed prefixes through  $M$ . Its upper endpoint  $E_+$  covers  $H_N$  and satisfies  $E_+ = g_+ + 2^{N-M}$ ; prefix adjacency is a separate transition hypothesis.

The results below have five different roles. The checked base feeds the ordinary one-row update. At a feedback row there is a dichotomy between a new full stage and a local gap. A separate recurrence advances such a gap when its coefficient inequalities hold. The quarter-band bound is only a necessary obstruction. Finally, the cofinal corollary is the sole step from arbitrarily deep full stages to half-membership.

**Proposition A.5** (verified base and one-row cylinder update). *The depth-18 selected window has  $R = 12$  and a suffix cylinder at cutoff  $M = 13$  with endpoint  $E = 17$ .*

*More generally, let a depth- $N$ , radius- $R$  selected window  $(a^{(k)})_{1 \leq k \leq R}$  have a suffix cylinder at cutoff  $M \leq N$  with endpoint  $E$ , where  $1 \leq N$  and  $R \leq E$ . Let  $1 \leq R' \leq H_{N+1}$ . If*

$$f_{a^{(k)}}(N+1) = C \quad (1 \leq k \leq R)$$

*and*

$$R' + C \leq 2R,$$

*then the radius- $R'$  successor remains a suffix cylinder, with endpoint*

$$E' = 2E - C.$$

The base and endpoint update are the [depth-18 cylinder](#) and [endpoint recurrence](#).

**Proposition A.6** (feedback-row dichotomy). *Let a full suffix-cylinder stage have depth  $N$ , cutoff  $M$ , with  $1 \leq N$ ,  $M+1 \leq N$ , and*

$$N+1 = 2(M+1).$$

*At depth  $N+1$ , either a full stage exists at cutoff  $M+1$ , or the exceptional carries form a two-sheet profiled gap stage. If  $27 \leq H_{N+1}$ , the residual alternative also realises a protected one-hole seam.*

Formalised by the [feedback dichotomy](#) and [two-family gap alternative](#).

**Proposition A.7** (profiled-gap transition). *Suppose a profiled gap stage has lower and upper next coefficients  $C_-$  and  $C_+$ , respectively. If*

$$\begin{aligned} 1 &\leq 2g_- - C_- - 1, & C_+ &\leq 2g_+, \\ 2g_- - C_- - 1 &\leq 2g_+ - C_+ \leq H_{N+1}, \\ H_{N+1} + C_+ &\leq 2H_N, \end{aligned}$$



then the stage advances one row with exact child gap

$$[g'_-, g'_+] = [2g_- - C_- - 1, 2g_+ - C_+].$$

At a cutoff boundary  $N = 2M + 1$ ,  $M \geq 4$ , the lower and upper prefixes acquire the forced bits 0 and 1; their consecutiveness is preserved. Under the displayed child-gap conditions at the next row, this promoted stage therefore advances with coefficients determined by its retained prefixes.

Formalised by the [two-coefficient gap recurrence](#) and [promote-and-advance theorem](#).

**Proposition A.8** (quarter-band swallow obstruction). *If a two-row update swallows a protected strip of bound  $B$ , and the lower coefficient pulse is at most the upper pulse plus 2, then*

$$B \leq 4(g_+ - g_- + 1) + 2. \quad (\text{A.1})$$

*This is a necessary quarter-band condition, not a contradiction.*

Formalised as the [swallow bound](#).

**Corollary A.9** (cofinal suffix-cylinder criterion). *If full suffix-cylinder stages exist at arbitrarily large depths, then there is an infinite set  $A \subseteq \mathbb{N}_{>0}$  such that*

$$\sum_{n \in A} \frac{1}{2^n - 1} = \frac{1}{2}.$$

This is the [positive-support cofinal-cylinder criterion](#).

**Status and relation to prior work.** The two transition propositions do not supply their coefficient inequalities, and the finite base does not supply the cofinal family required by Corollary A.9. The strict-tail achievement-set literature supplies the surrounding Cantor-set geometry [16], not this finite carry recurrence; the comparison record makes no novelty claim for the recurrence itself.

#### A.5.4 Verified finite suffix-cylinder stages

The finite record has three milestones.

- (1) *Depth 27.* Before the first half-divisor feedback row, a full selected window exists at every depth from 18 through 27, and the depth-27 endpoint covers the whole protected strip: [stages through depth 27](#) and [depth-27 strip coverage](#).
- (2) *Depth 29.* The threshold-stage continuation still covers the strip and has a common suffix cylinder at every cutoff from 14 through 25, while cutoff 26 is explicitly not common: [depth-29 strip coverage](#) and [cutoff-26 obstruction](#).
- (3) *Depth 52.* An independent exact run reaches depth 51 with endpoint 51,327,745, beyond the next feedback head threshold, and the promoted successor at depth 52 is checked: [depth-51 threshold margin](#) and [depth-52 terminal-strip witness](#).

These are finite milestones, not an unbounded induction; they do not provide the arbitrarily large stages required by Corollary A.9.

#### A.5.5 Half-divisor unit drop

At the doubled feedback row, changing only the terminal half-divisor bit from 0 to 1 increases the support coefficient by exactly one: [the terminal-bit unit-drop identity](#). The same holds for any explicitly identified adjacent boundary pair, [the boundary-pair unit-drop identity](#). The inverse-parent recurrence transfers such a pair to the next protected seam while retaining the exact unit-drop witness, [the rewind unit-drop transfer](#) and [the protected-seam transfer](#). This is the local coefficient input for the rewind-seam profile, not a global construction of such boundary pairs.

### A.5.6 A finite rewind-boundary obstruction

The scalar one-row rewind data alone still does not supply that missing boundary predicate. Lean gives a concrete depth-26 selected two-word window whose depth-27 explicit step has rewind history [1], is a width-two seam pair, and satisfies the doubled-row identity, but has no depth-13 half-divisor boundary-pair witness: [the depth-27 boundary-pair counterexample](#). Thus the protected-window buffer in the compactness criterion is a genuine hypothesis, not a consequence of the scalar seam facts alone.

### A.5.7 Explicit carry coordinates at the final non-right transition

For every adjacent cut at rank  $s \geq 5$ , the transition carry of the terminal-augmented below support equals the concrete middle coordinate:

$$4 \text{ remainder} - \text{belowPulse} - 4,$$

and identifies the terminal-augmented above support with the negative upper coordinate

$$-(4 \text{ overshoot} + \text{abovePulse} + 4).$$

The generic row-word identity is [row-word transition identity](#), with the below and above specialisations at [middle-coordinate identity](#) and [upper-coordinate identity](#). These identities make the remaining quantitative task exact: they do not prove exponential lower bounds, sign persistence, or a universal escape mechanism.

### A.5.8 Incompatibility with Campbell’s prime-progression bound

Crandall asked whether the binary block 11 occurs infinitely often in the Erdős–Borwein constant; Campbell proved that it does using primes in an arithmetic progression whose modulus  $d$  is constrained by  $d^4 \leq X$ , where  $X$  is the prime-search bound [18, 19]. A proposed route combines this input with the finite Mersenne shift from a putative last greedy skip. If  $t$  is the unavoidable phase factor and  $P$  the prime cap forced by the fatal window, the required inequalities are  $2 \leq d$ ,  $t \leq d$ ,  $P \leq 4t$ , and  $d^4 \leq P$ ; they are inconsistent. Formalised: [the period-freeze incompatibility](#) and [the parameter incompatibility](#).

The associated shifted-zero condition is not an independent hypothesis. It is exactly equivalent both to infinitely many actual greedy skips and to  $1/2 \in \mathcal{A}$ : [the shifted-zero/skip equivalence](#) and [the shifted-zero/half-membership equivalence](#). This settles one synchronisation question and gives an equivalent reformulation; it does not decide the half-membership question stated above.

**Appendix consequence.** This appendix supplies the local implications behind Theorems 4.11 and 4.14, Proposition 4.15, and Corollary 4.16. It does not establish the corollary’s global hypothesis. A sufficient hypothesis is either a cofinal window, seam, or terminal approximation producing  $1/2 \in \mathcal{A}$ , or both an exclusion of every actual middle transition with  $C_D = -2, -1$  and a proof of  $\Theta_D < C_D$  at every remaining non- $(-3)$  middle transition. The bounded checks and local no-go theorems provide neither route.

## B Auxiliary rationality criteria from binary carry systems

This appendix supplies the general results assembled in Theorem 4.4 and Corollaries 4.5–4.8, and the detailed greedy geometry cited after Theorem 4.3. Its new working objects are a tempered integer tail orbit, its Boolean Möbius coordinate, the wrap-excess decomposition modulo the

odd denominator, and exact rational death certificates for the achievement set. Appendices B.1–B.4 describe necessary or equivalent forms of a hypothetical rational support; Appendix B.5 describes the ambient achievement set and its one-sided finite nonmembership test. None of these objects by itself excludes a rational value.

The reading order is: a generic tail-orbit rationality normal form in Appendix B.1; its Boolean–Möbius specialisation in Appendix B.2; divisor-coverage constraints in Appendix B.3; denominator-wrap, reciprocal-mass, and unbounded-state constraints in Appendix B.4; and the exact greedy geometry with one-sided finite death certificates in Appendix B.5.

The dependency chain is

$$\begin{array}{ll}
\text{a rational support value} & \iff \text{a tempered integral tail orbit (B.1),} \\
\text{a specified support fraction } p/q & \iff \text{a Boolean–Möbius carry certificate (B.2),} \\
\text{a nonempty rational support} & \implies \text{sublogarithmic zero gaps (B.3),} \\
\text{a nonempty support, odd part } v > 1 & \implies \text{wrap and reciprocal-mass bounds (B.4),} \\
\text{an infinite rational support} & \implies \text{unbounded carry states (B.4),} \\
\text{an infinite dyadic-denominator support} & \implies \text{the reciprocal-mass alternative (B.4).}
\end{array}$$

The last four lines list independent necessary conditions, at the weakest hypotheses under which they are proved, not converses and not contradictions. Appendix B.5 is orthogonal to this rationality chain: finite greedy death proves nonmembership in the achievement set, whereas survival through any finite depth proves nothing.

The prior-art boundary is theorem-family-specific. The integral carry recurrence has a direct antecedent in Wang [17]; the strict-tail geometry is recorded by Kovač–Tao [16]; Möbius inversion, repetend algebra, and divisor averaging are classical (see, for example, Apostol [6]). We make no claim of mathematical priority for the converse/rigidity, certificate-normal-form, or coupled reciprocal-mass/collision statements collected here.

## B.1 Exact normal form: tail-orbit rigidity

This subsection supplies the generic equivalence used in Appendix B.2. It converts rationality into an integer recurrence with a boundary condition; it does not itself exclude such an orbit or impose the Boolean support constraint.

For a coefficient sequence  $\gamma : \mathbb{N} \rightarrow \mathbb{N}$  with  $\gamma(n) \leq n$ , write

$$X_\gamma = \sum_{n \geq 1} \frac{\gamma(n)}{2^n}, \quad T_\gamma(N) = \sum_{j \geq 1} \frac{\gamma(N+j)}{2^j},$$

the series and its scaled tail after  $N$  binary digits. Call an integer sequence  $u$  a *tempered orbit* for  $\gamma$  with multiplier  $v \geq 1$  if

$$u(N+1) = 2u(N) - v\gamma(N+1) \text{ for all } N, \quad \text{and} \quad u(N)/2^N \rightarrow 0.$$

The recurrence is exact binary long division against the coefficient stream; the boundary condition forbids the homogeneous solution. Wang uses the integral carry recurrence forced by rationality for the weighted binary special case  $\gamma(n) = nd_n$  [17]. The theorem below extends that forward mechanism to arbitrary nonnegative  $\gamma(n) \leq n$ ; its explicit converse and subexponential-orbit rigidity are the local additions, with no priority claim attached. Wang’s positive-density theorem and its Erdős #260 corollary are not used or formalised here.

**Theorem B.1** (tail-orbit rigidity). *Let  $\gamma(n) \leq n$  for all  $n$ . Every tempered orbit is the scaled tail,  $u(N) = vT_\gamma(N)$  for all  $N$ ; and  $X_\gamma$  is rational if and only if a tempered orbit exists for some multiplier  $v \geq 1$ .*

Formalised: [the scaled-tail orbit identity](#) (rigidity) and [the rationality–orbit equivalence](#) (the criterion), restated in Mathlib’s irrationality vocabulary at [the non-irrationality–orbit equivalence](#). The engine is one observation: a real orbit with  $d(N+1) = 2d(N)$  and  $d(N) = o(2^N)$  vanishes identically ([zero-tail rigidity](#)).

The tempered boundary is essential: the homogeneous parasite  $N \mapsto 2^N$  can be added to any orbit without disturbing the recurrence, so positivity of an orbit certifies nothing by itself, and positivity is nowhere advertised as an equivalent criterion. The linear-growth hypothesis covers both constants introduced in Section 1:  $\varphi(n) \leq n$  for the #249 coefficients, and  $f_A(n) \leq \tau(n) \leq n$  for the #257 support coefficients  $f_A(n) = \#\{a \in A : a \mid n\}$ . The results below instantiate the criterion on supports. Its separate totient anti-compression consequence belongs to the #249 rank route in Appendix C.4.

## B.2 Exact #257 coordinates: Boolean–Möbius carry certificates

This subsection has one preliminary boundary result and then one exact equivalence chain. The boundary result rules out extracting the desired Boolean support directly from the signs in  $L(\mu) = 1/2$ ; it is not a premise of the carry equivalence that follows.

**Proposition B.2** (Möbius-sign support no-go). *Let*

$$A_- = \{d \geq 2 : \mu(d) = -1\}, \quad A_+ = \{d \geq 2 : \mu(d) = 1\}.$$

*Then*

$$\sum_{d \in A_-} \frac{1}{2^d - 1} = \frac{1}{2} + \sum_{d \in A_+} \frac{1}{2^d - 1} \geq \frac{1}{2} + \frac{1}{63} > \frac{1}{2}.$$

*In particular, the signed identity  $L(\mu) = 1/2$  cannot be converted into a Boolean support of value  $1/2$  merely by selecting the negative Möbius indices.*

Formalised: [the negative-Möbius decomposition](#) and [the strict lower bound](#). The term  $d = 6$  supplies the displayed  $1/63$ . This rules out one sign-truncation route only; it does not exclude another infinite Boolean support with value  $1/2$ .

**The exact coordinate chain.** At base 2 the support series of Section 4 is itself a binary coefficient series,  $\sum_{n \in A} 1/(2^n - 1) = X_{f_A}$  ([the support-series/binary-series identity](#)). Boolean Möbius inversion first identifies the support from its divisor coefficients; tail-orbit rigidity then identifies rationality with a tempered integer orbit. Theorem B.4 is exactly the composition of those two equivalences. The divisor transform and Möbius inversion are classical Lambert-series tools [6, 7, 8]; the contribution here is their Lean-checked composition with the tempered-orbit criterion.

**Proposition B.3** (Boolean Möbius inversion, both directions). *On positive integers,  $f_A = \mathbf{1}_A * \mathbf{1}$  and  $\mu * f_A = \mathbf{1}_A$ , where  $\mathbf{1}_A$  is the indicator of  $A$ . Conversely, every integer-valued arithmetic function  $f$  whose Möbius transform is Boolean,  $(\mu * f)(n) \in \{0, 1\}$  for all  $n \geq 1$ , satisfies  $f = f_B$  for the support  $B = \{n \geq 1 : (\mu * f)(n) = 1\}$  selected by that transform.*

Formalised: [the positive-support divisor identity](#), [the Möbius inversion identity](#), and, with no search or finiteness hypothesis, the converse [the Boolean reconstruction theorem](#).

On the carry side, Theorem B.1 applies at  $\gamma = f_A$ : the support series is rational exactly when a tempered carry orbit for  $f_A$  exists ([the rationality–carry equivalence](#)); a displayed fraction  $p/q$  corresponds to an orbit  $U$  with multiplier  $q$  and initial value  $U(0) = p$  ([the support-fraction carry equivalence](#)); and exact division recovers the coefficient from the orbit,  $(2U(N) - U(N+1))/q = f_A(N+1)$  ([carry-quotient recovery](#)). The elementary divisor-pair bound  $\tau(n) \leq 2\lfloor\sqrt{n}\rfloor$  confines the tails to a square-root strip,  $T_{f_A}(N) \leq 2\sqrt{N} + 4$ . For a nonempty support the orbit states

are positive and satisfy  $U(N) \leq q(2\sqrt{N} + 4)$  (the carry upper bound); and the strip is strong enough to re-derive the tempered boundary, so the analytic side condition can be traded for one inequality. The two coordinate changes compose.

**Theorem B.4** (certificate-level equivalence). *Fix an integer  $p$  and an integer  $q \geq 1$ . There is a nonempty support  $A$  of positive integers with  $\sum_{n \in A} 1/(2^n - 1) = p/q$  if and only if there is an integer sequence  $U$  with  $U(0) = p$  such that: every  $U(N)$  is positive;  $U(N) \leq q(2\sqrt{N} + 4)$ ;  $q$  divides  $2U(N) - U(N + 1)$  for every  $N$ ; and the Möbius transform of the quotient sequence  $(2U(N) - U(N + 1))/q$  is Boolean. The support is not guessed: it is reconstructed from the certificate by Proposition B.3.*

Formalised as the normalised support-fraction certificate equivalence.

**Example B.5** (the support  $\{2, 3\}$ ). The support  $\{2, 3\}$  has value  $\frac{1}{3} + \frac{1}{7} = \frac{10}{21}$  (the two-point series value), its orbit is the pure period-six cycle 10, 20, 19, 17, 13, 26 with multiplier 21 (the tempered two-point orbit), and the Möbius transform of the carry quotient recovers exactly  $\{2, 3\}$  (support recovery).

These are coordinates, not by themselves a strategy: nothing here excludes infinite Boolean carry paths, and no finite search is promoted to a completeness argument.

### B.3 Necessary condition I: sublogarithmic divisor coverage

This theorem is a one-way consequence of a rational support value and is used in Corollary 4.6. It limits zero gaps in the divisor-coefficient stream; it neither reconstructs a support nor contradicts the existence of an infinite rational support.

The coefficient  $f_A(n)$  counts the elements of  $A$  dividing  $n$ . Say that a zero window of length  $h$  starts after  $c + N$  when  $f_A(c + N + j + 1) = 0$  for every  $0 \leq j < h$ . Thus no integer in that interval is divisible by an element of the support.

**Theorem B.6** (sublogarithmic zero windows). *Let  $A$  contain a positive integer and suppose*

$$\sum_{a \in A} \frac{1}{2^a - 1} = \frac{p}{2^c v}, \quad p \in \mathbb{Z}, \quad c \in \mathbb{N}, \quad v \geq 1.$$

*For every  $\varepsilon > 0$  there is a constant  $B \geq 0$ , depending only on  $\varepsilon, c$ , and  $v$ , such that every zero window after  $c + N$  satisfies*

$$h \leq \varepsilon \log_2(N + 1) + B, \quad \log_2 x := \frac{\log x}{\log 2}.$$

*The bound is uniform in  $A, p, N$ , and  $h$ .*

Formalised: sublogarithmic zero-window bound.

The proof first establishes, for each integer  $k \geq 2$ , the explicit divisor estimate

$$\tau(n)^k \leq (k^{2^k})^k n.$$

It transports the resulting  $n^{1/k}$  bound through the binary tail and then uses the exact carry recurrence across a zero window. Raising the resulting inequality to the  $k$ th power absorbs the occurrence of  $h$  on the right and leaves a coefficient  $1/(k - 1)$  in front of the binary logarithm. Choosing  $k$  after  $\varepsilon$  gives the theorem. This controls gaps in divisor coverage; it does not assert a subpower bound for the carry state itself.

For one fixed support the conclusion is elementary, since every zero window is shorter than the least support element. The content is uniformity:  $B$  does not depend on  $A$ , and taking  $N = 0$  shows that a rational value with denominator data  $(c, v)$  bounds the least element of its support in terms of  $(\varepsilon, c, v)$  alone. Against the half-value fibre the constraint is inert: the greedy candidate support there contains 2, so its divisor stream has no zero window of length above one and satisfies the bound with room to spare.

## B.4 Necessary conditions II: wraps, mass, and unbounded states

The dependency order is

doubling residues  $\longrightarrow$  wrap count  $\longrightarrow$  reciprocal-mass bound  $\longrightarrow$  collision strengthening.

The final unbounded-state theorem uses the same common-multiple collision mechanism without a summability hypothesis. The one-wrap classification identifies the extremal residue cycles; it is a side branch, not a premise of the general mass bound.

Suppose now the support series equals  $p/(2^c v)$  with  $v$  odd and  $A$  containing a positive element. Let  $U$  be the unshifted tempered orbit of Appendix B.2 with multiplier  $q = 2^c v$ . The odd-denominator coordinate used here is

$$u(N) = \frac{U(c+N)}{2^c} = v T_{f_A}(c+N);$$

this is the state denoted  $u_N$  in Theorem 4.4. Rationality makes  $u(N)$  a positive integer satisfying  $u(N+1) + v f_A(c+N+1) = 2u(N)$ . Moreover,  $u(N)$  reduces modulo  $v$  to the doubling residue  $\bar{p}_N = p 2^N \bmod v$  ([shifted natural-state existence](#)). Doubling a least residue either stays under  $v$  or wraps past it exactly once, so each step emits a wrap digit  $k_N = \lfloor 2\bar{p}_N/v \rfloor \in \{0, 1\}$ , and over any cycle length  $h$  with  $2^h \equiv 1 \pmod{v}$  the repetend identity holds:

$$\sum_{N < h} \bar{p}_N = v \cdot w, \quad w = \sum_{N < h} k_N,$$

with  $w \geq 1$  when  $p$  is coprime to  $v > 1$ . The formal development records the displayed equality as the [wrap-count identity](#) and its non-vanishing as the [positive-wrap theorem](#).

**Proposition B.7** (one-wrap classification). *Let  $v > 1$  be odd,  $p$  coprime to  $v$ , and  $h \geq 1$  with  $2^h \equiv 1 \pmod{v}$ . If the cycle of  $p$  wraps exactly once, then  $v = 2^h - 1$  and the starting residue is a power of two:  $\bar{p}_0 = 2^a$  for some  $a < h$ .*

The one-wrap cycles are thus exactly the Mersenne repetends  $2^a/(2^h - 1)$ . Formalised: [one-wrap classification](#), from the generic closed-cycle form [one-wrap cycle classification](#), and instantiated at the true order  $h = \text{ord}_v(2)$  ([odd-order one-wrap classification](#)). The classification is algebraic; a finite check of 446 coprime starting residues across twelve modulus/order rows is retained as kernel-checked validation, not as the proof ([446-start finite validation](#)).

The analytic bridge is Cesàro averaging. Write  $\rho(A) = \sum_{a \in A} 1/a$  for the reciprocal mass. When the reciprocal family is summable, the mean of  $f_A(1), \dots, f_A(N)$  converges to  $\rho(A)$  — the density of the multiples of  $a$ , summed over the support — and the mean of the scaled tails has the same limit ([divisor-mean limit](#), [coefficient-tail mean limit](#)). The divergent case is carried as an explicit disjunction, not through a default value assigned to a divergent formal sum. Each tail state splits exactly as  $u(N) = v e_N + \bar{p}_N$  with integral excess  $e_N = \lfloor u(N)/v \rfloor \geq 0$ .

**Proposition B.8** (order-wrap lower bound, with exact excess mean). *Let  $A \subseteq \mathbb{N}_{>0}$  be nonempty and suppose*

$$X_A = \frac{p}{2^c v}, \quad p \in \mathbb{Z}, \quad c \in \mathbb{N}, \quad v > 1 \text{ odd}.$$

*Let  $h = \text{ord}_v(2)$ , let  $w$  be the number of wraps in the least-residue cycle  $p 2^N \bmod v$  for  $0 \leq N < h$ , and set*

$$u(N) = v T_{f_A}(c+N), \quad e_N = \left\lfloor \frac{u(N)}{v} \right\rfloor.$$

*Then  $\sum_{a \in A} 1/a$  diverges or  $\rho(A) \geq w/h$ ; if moreover  $p$  is coprime to  $v$ , then  $w \geq 1$ , so  $\rho(A) \geq 1/\text{ord}_v(2)$ . In the summable case the bound is the periodic part of an exact identity: the Cesàro mean of  $e_N$  converges, and*

$$\rho(A) = \frac{w}{h} + \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n < N} e_n.$$



The divergent-or-bounded conclusion is formalised by the [wrap-ratio mass alternative](#), its co-prime specialisation by the [odd-order mass alternative](#), and the automatic excess limit in the fraction-facing coordinates above by the [shifted excess-mean identity](#). A rational value thus ties the odd part of its denominator to the sparsity of its support: a summable support with small  $\rho(A)$  can only display denominators whose odd part has small multiplicative order of 2.

**Corollary B.9** (collision strengthening at common multiples). *In the summable case, let  $F \subseteq A$  be finite and let  $L \geq 1$  be a common multiple of  $F$ . Then*

$$\rho(A) \geq \frac{w}{h} + \frac{\lceil (|F| - 1)/2 \rceil}{L}.$$

*In particular, an infinite support whose value is a dyadic rational  $p/2^c$  has  $\sum_{a \in A} 1/a$  divergent or  $\rho(A) > 1$ .*

The mechanism is a collision: at every positive multiple of  $L$  at least  $|F|$  support divisors land on the same index, so  $f_A \geq |F|$  there, while the exact carry equation  $f_A(n) + e_n = k_{n-1} + 2e_{n-1}$  with  $k_{n-1} \leq 1$  lets one step absorb at most one unit — so the preceding excess must spike to at least  $\lceil (|F| - 1)/2 \rceil$ , and the spikes recur with density  $1/L$ . The three steps are formalised by the [shifted common-multiple bound](#), [shifted wrap-excess recurrence](#), and [dyadic reciprocal-mass alternative](#).

**Theorem B.10** (carry states are unbounded over infinite supports). *Let  $A$  be infinite with value  $p/(2^c v)$ ,  $v \geq 1$ . Then the positive integer carry state  $u(N) = v T_{f_A}(c + N)$  is unbounded: for every  $B$  there is an  $N$  with  $u(N) > B$ .*

The proof picks  $2B + 1$  positive support elements; at a common multiple  $L$  of all of them, the local inequality  $1 + v|F| \leq 2u(L - c - 1)$  pushes the state past  $B$ . The conclusion is the [fraction-facing unbounded-state theorem](#); its local engine is the [support-cardinality state bound](#). So a rational value over an infinite support cannot run on a bounded carry alphabet. That rules out finite-state readings of the carry system; it does not decide the problem. Nothing in the constraints above prevents an infinite support from satisfying every one of these constraints. The repetend identities and divisor averages used here are classical. We make no claim of mathematical novelty for the coupled reciprocal-mass bounds, collision strengthening, or global unboundedness statement.

## B.5 Independent geometry and one-sided nonmembership certificates

Finally, consider the set of all values a base-2 support series can take. For  $n \geq 1$  put  $w_n = 1/(2^n - 1)$ , let  $T(n)$  be the mass strictly after exponent  $n$ , and let

$$\mathcal{A} = \left\{ \sum_{n \in A} w_n : A \subseteq \mathbb{N}, 0 \notin A \right\},$$

the *Mersenne achievement set*, the exponent 0 being normalised away as analytically invisible. Coding by supports of positive exponents is literally the formal support series ([support-value identity](#)). In this language the base-2 #257 statement reads: the rational members of  $\mathcal{A}$  are exactly the finite subset sums. Nothing below decides that. Kovač and Tao verify, for every fixed integer  $t \geq 2$ , that

$$\sum_{\ell > n} \frac{1}{t^\ell - 1} < \frac{1}{t^n - 1},$$

and deduce distinct Lambert subsums and a Cantor set [16]. At  $t = 2$  this is the relevant Lambert-series instance of Kakeya's classical strict-tail criterion for subsum sets [9]. The local contribution is the checked greedy criterion, quantitative enclosure, and finite nonmembership certificates over that known geometry; it does not attribute those refinements to the cited sources or settle #257.



**Theorem B.11** (superincreasing geometry and greedy membership). *For every  $n \geq 1$  the weight dominates the whole tail after it,  $T(n) < w_n$ , with the two-scale gap  $w_n - T(n) = \frac{2}{3}4^{-n} + O(8^{-n})$  and explicit valid  $O$ -constant 3. Consequently a real  $x$  belongs to  $\mathcal{A}$  exactly when  $x \geq 0$  and every greedy residual of  $x$  is at most the remaining tail; and normalised support coding is injective, so each member of  $\mathcal{A}$  has a unique support, which the greedy recursion recovers bit by bit.*

Formalised: [strict-tail inequality](#), [gap big-O estimate](#) (explicit bound [explicit gap bound](#)), [greedy survival criterion](#), and [normalised support-value injectivity](#), transferred to the kernel series at [normalised support-series injectivity](#). The nonnegativity guard in the membership criterion is necessary: without it a negative target would survive every level while lying outside  $\mathcal{A}$ .

The binary coding also permits the global geometry to be checked directly.

**Proposition B.12** (fat-Cantor geometry). *The achievement set  $\mathcal{A}$  is compact, perfect, totally disconnected, and nowhere dense. Its Lebesgue measure is exactly one.*

Formalised: [measure-one theorem](#), [perfectness](#), [total disconnectedness](#), and [nowhere density](#). Kovač–Tao supply the strict-tail Cantor-set context cited above. The formalisation records these exact properties without making a novelty or priority claim for them.

**Proposition B.13** (exact rational runs and finite nonmembership certificates). *The greedy recursion executed in exact rational arithmetic agrees step by step with the real recursion under casting. For a rational target  $x$ , write  $r_n(x)$  for its exact rational residual after level  $n$  and, for a lookahead  $\ell \geq 0$ , put*

$$\hat{T}_{n,\ell} = \sum_{k=0}^{\ell-1} w_{n+k+1} + 2w_{n+\ell+1} \in \mathbb{Q}.$$

*The sum is empty when  $\ell = 0$ , and always  $T(n) < \hat{T}_{n,\ell}$ . Hence the finite inequality  $\hat{T}_{n,\ell} \leq r_n(x)$  soundly proves  $x \notin \mathcal{A}$ ; its level-one instance shows  $3/4 \notin \mathcal{A}$ . A rational number already known to lie in  $\mathcal{A}$  has computable support bits.*

Formalised: [rational–real greedy agreement](#), [rational tail-enclosure bound](#), [finite death-certificate soundness](#), [three-quarters nonmembership](#), and [rational-member support-bit equivalence](#). The certificates are one-sided: survival through any finite depth proves nothing about membership, and no decidability of membership in  $\mathcal{A}$  is asserted. Neither the global geometry nor the finite death certificates exclude rational values for either open series.

**Appendix consequence.** The transfer to the main text is Theorem 4.4 followed by the four independent Corollaries 4.5–4.8. Rationality yields the exact carry recurrence; any nonempty rational support must satisfy the sublogarithmic divisor-coverage bound, and the odd-part wrap and mass bounds when its denominator has odd part  $v > 1$ ; an infinite one must in addition satisfy the unbounded-state theorem and, at a dyadic denominator, the reciprocal-mass alternative. Appendix B.5 also supplies the finite death certificates used in Theorem 4.3. What is not proved is that the support attached to either target violates one of these necessary conditions. That contradiction, not another rational normal form, is the missing premise.

## C Technical reductions for the totient-certificate branch

This appendix expands the supporting mathematics behind Proposition 5.4, Theorems 5.5 and 5.6, and Theorem 5.7. It contains exact cone-flatness and target-interval reformulations, finite residue projections, denominator and bit-lifting formulae, bounded certificate families, and rigorous obstructions to several candidate strategies. Each statement retains its own hypotheses and status. None proves the unbounded certificate supply (5.1).

The exact chain is

$$\begin{aligned} R_{N+h} - R_N \notin \mathbb{Z} &\iff \exists L \geq 1, \text{ Sep}(h, N, L), \\ S \notin \mathbb{Q} &\iff \forall h \geq 1 \forall N_0 \exists N \geq N_0 \exists L \geq 1, \text{ Sep}(h, N, L). \end{aligned}$$

Appendix C.1 gives equivalent coordinates for this fixed-to-cofinal reduction. Appendix C.2 distinguishes one-scale mechanisms from the conditional cofinal implications that use them. Appendices C.3–C.4 expose possible inputs to an unbounded supply theorem and rule out several shortcuts. None of the latter three sections proves the missing cofinal supply in the second line.

### C.1 Exact cofinal normal forms and cone consequences

This subsection contains two different kinds of statement. Cone flatness is a necessary consequence of rationality that a finite joint refuter may contradict at chosen vertices. The diagonal-pincer identity and its cofinal full-target-avoidance corollary are the exact fixed-scale and global normal forms of the open supply. Only the latter corollary is equivalent to the irrationality of  $S$ .

**Cone flatness and completeness.** On the cone  $\{kM_t : k \geq 1\}$ , rationality forces a single fractional constant across the whole cone.

**Proposition C.1** (cone flatness). *If  $S$  is rational, then for every sufficiently large  $t$  there is  $\theta_t \in [0, 1)$  such that the fractional part of the tail  $R_{kM_t}$  equals  $\theta_t$  for every  $k \geq 1$ .*

Formalised: [cone flatness under rationality](#).

Proposition 5.4 states fixed-parameter certificate completeness, and Theorem 5.5 gives its global pointwise form. Their additional role here is cone-level: a joint finite refuter can interrogate several cone vertices at once and is sharper than any pairwise test. It refutes the single-fractional-part model that rationality predicts on a finite sample ([finite cone-nonflatness refuter](#)). This finite cone predicate is sound in the direction just stated: a certificate produces a non-integral pair and hence contradicts the sampled flatness model. No converse is proved saying that every non-flat finite sample, or every irrational value of  $S$ , must be detected by this joint predicate. It is therefore a refuter, not a complete decision procedure.

**An exact target interval.** The diagonal calculation also admits a more local normal form. Put  $H_t = M_t$  and

$$D_t = R_{2H_t} - R_{H_t}.$$

Separate an exact rational Möbius shadow  $Q_t = a_t/d_t$  in lowest terms and define the remaining *foreign defect* by  $F_t = D_t - Q_t$ ; the shadow collects the explicitly summed Möbius channels, and the defect is everything that explicit part does not see. The full target is the residue class of  $-a_t$  modulo  $d_t$ :

$$\text{Hit}(t) \iff \exists k \in \mathbb{Z}, \quad d_t F_t = -a_t + d_t k.$$

Since  $D_t = a_t/d_t + F_t$ , this is exactly the condition  $D_t \in \mathbb{Z}$ ; no complementary term is discarded or assigned a conjectural sign.

**Proposition C.2** (diagonal pincer decomposition). *For every  $t$ , diagonal integrality is equivalent to the foreign diagonal defect meeting the full target.*

This is the [full-target equivalence](#).

**Corollary C.3** (exact full-target avoidance characterisation). *The following are equivalent:*

$$\begin{aligned} S \notin \mathbb{Q}; \\ \forall t_0 \in \mathbb{N} \exists t \geq t_0, \quad \neg \text{Hit}(t). \end{aligned}$$

The reverse implication is the [unbounded target-avoidance criterion](#). For the forward implication, Theorem 5.7 supplies cofinally many non-integral diagonal differences, and Proposition C.2 converts each one into a missed full target. Thus the proposition is the fixed-scale equivalence and the corollary is its exact global form. The cofinal avoidance statement remains open.

## C.2 Fixed-scale mechanisms and conditional cofinal implications

This subsection is organised by quantifier. Its enclosures, predicates, checked certificate families, and projection equivalences are finite or one-scale mechanisms. Whenever a corollary concludes irrationality, its additional premise is an unproved cofinal supply of the relevant finite mechanism. No result here constructs such a supply or fills the missing quantifier in Appendix C.1.

**A one-way analytic enclosure.** Truncate the Möbius-square identity of Proposition D.6 after  $D$  terms. The resulting rational quantity gives a Lambert-projected centre  $C_{H,D}$  for the scaled diagonal difference. The error is exactly a signed squared-Mersenne tail:

$$\text{scaledDiagonal}(H) - C_{H,D} = c_H \sum_{d>D} \frac{\mu(d)}{(2^d - 1)^2},$$

where  $c_H$  is the explicit diagonal coefficient in the formal source. The absolute value of the sum is bounded geometrically by  $4/[3(2^{D+1} - 1)^2]$ . Thus a rational separation of the centre from every integer by more than the scaled bound proves target avoidance; the argument does not estimate the sign of the tail.

**Proposition C.4** (squared-Mersenne enclosure). *The scaled diagonal tail difference differs from its Lambert-projected centre by the squared-Mersenne tail. Separation of the centre from the full target by more than the proved tail bound forces target avoidance.*

Formalised: [squared-Mersenne tail identity](#) and [Lambert-projected separation criterion](#). The enclosure is one-way: separation at a chosen truncation forces target avoidance, but target avoidance is not asserted to force that separation.

**A reduced-denominator finite predicate.** Separately, there is a stricter finite condition for an odd reduced denominator  $u$ . Its predicate  $\text{UnitGap}(u, N, K)$  asks that every candidate integer in the associated dyadic tail interval be non-coprime to  $u$ . At a prime power  $u = p^e$ , the condition is equivalent to the exact alternative that the interval has no candidate, or that it has one candidate and  $p$  divides that candidate. Thus such an interval has at most one candidate.

**Proposition C.5** (prime-power unit-gap ceiling). *For every prime power  $p^e$  with  $e > 0$ , a  $\text{UnitGap}(p^e, N, K)$  has candidate count at most one. The row  $(u, N, K) = (3, 3, 5)$  satisfies this predicate, although its corresponding ordinary empty-gap inequality fails.*

Formalised: [prime-power unit-gap characterisation](#), [prime-power one-candidate bound](#), [three-window unit-gap certificate](#), and [three-window ordinary-gap failure](#). This is a finite certificate-level strengthening. We do not establish that rationality of  $S$  supplies these predicates at unbounded parameters.

**The certificates are non-empty.** The initial segment of the required supply is machine-checked, so the reduction is not vacuous. By completeness, each of these is a kernel-checked proof of a specific non-integral tail difference, not a numerical experiment.

**Example C.6** (verified finite instances, expanded record). The following are checked by the Lean kernel:

- $\text{Sep}(h, 12, 16)$  for every period  $h \in [1, 8]$ ;
- finite certificates for every period  $h \in [1, 16]$  at a fixed window;
- diagonal certificates  $\text{Sep}(M_t, M_t, L)$  at  $t = 1$  and the 27 strict lcm-jump scales through 64, namely
 
$$t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59, 61, 64\};$$
- a tabulated joint cone-vertex certificate.

The four validation classes are formalised, in order, by the [small-window certificate family](#), [periods through sixteen](#), [28 imported diagonal certificates through scale 64](#), and [cone-cell certificate table](#). These instances show that each layer of the reduction is non-vacuous and, by completeness, exact. They are not evidence that the unbounded supply exists.

The first row of this table is worked in full following Example 5.8. The soundness theorem ([certificate soundness](#)) converts each successful row into one exact non-integral tail difference; no row asserts that such differences occur at unbounded scales.

**Finite foreign-residue projection.** There is a second finite presentation of the same obstruction. For a cutoff  $D$ , split the finite Möbius residue diagonal into the channels  $d \leq D$  which divide  $H$  and the remaining channels. The divisor part is exactly the truncated Möbius shadow, while every retained nondivisor channel is explicit. Above the stable cutoff  $2H$ , a finite window of omitted channels is bounded by

$$c_H \left( \frac{2}{2^D} + \frac{4}{3 \cdot 4^D} \right),$$

where  $c_H$  is the diagonal coefficient. Thus, if the actual foreign defect agrees with this projection within the displayed bound, and the projected quantity is farther than that bound from every integer, then the full target is missed.

**Proposition C.7** (finite foreign-residue projection). *Let  $F_H^{(D)}$  be the retained nondivisor projection at cutoff  $D$ , let  $F_H$  be the actual foreign defect, and let  $Q_H$  be the scaled explicit divisor shadow. The finite residue diagonal is exactly the sum of  $F_H^{(D)}$  and its divisor-channel complement. For  $2H \leq D$ , put*

$$B_{H,D} = c_H \left( \frac{2}{2^D} + \frac{4}{3 \cdot 4^D} \right).$$

If

$$|F_H - F_H^{(D)}| \leq B_{H,D} \quad \text{and} \quad |Q_H + F_H^{(D)} - z| > B_{H,D} \quad \text{for every } z \in \mathbb{Z},$$

then the full target at height  $H$  is not attained.

Formalised: [foreign-divisor channel split](#), [omitted-channel bound](#), and [projected-separation criterion](#). The passage from finite windows to the analytic foreign complement, including the corresponding kernel identity, is not supplied here. Nor is an unbounded family of separated projections; these are parts of the open certificate-supply problem.

Second, the rational Möbius shadow has an exact reduced denominator. If  $r_t$  is the square-free kernel of  $H_t$ ,  $s_t = H_t/r_t$ , and  $J(r_t) = \prod_q (q^2 - 1)$  is the odd Jordan scalar, the product running over the odd prime divisors  $q$  of  $r_t$ , then

$$d_t = \frac{2^{r_t} - 1}{\gcd(2^{r_t} - 1, s_t |J(r_t)|)}.$$

For  $t \geq 5$ , the product of the Mersenne factors  $2^p - 1$  over primes  $t/2 < p \leq t$  divides  $d_t$ ; in particular  $d_t \geq 2^{\lfloor t/2 \rfloor}$ . This is a denominator theorem for  $Q_t$  only. Cancellation by  $F_t$  is exactly what the full-target condition retains.

**Proposition C.8** (Mersenne-shadow denominator). *Let  $r_t$  be the square-free kernel of  $H_t$ , let  $s_t = H_t/r_t$ , and let  $J(r_t)$  be the odd Jordan scalar. The reduced denominator  $d_t$  of the scaled Möbius shadow is*

$$d_t = \frac{2^{r_t} - 1}{\gcd(2^{r_t} - 1, s_t |J(r_t)|)}.$$

If  $t \geq 5$ , then

$$\prod_{\substack{p \text{ prime} \\ t/2 < p \leq t}} (2^p - 1) \mid d_t, \quad d_t \geq 2^{\lfloor t/2 \rfloor}.$$

Formalised: [upper-half Mersenne divisor bound](#) and [exact shadow denominator](#).

Finally, the remaining diagonal error can be represented by a finite integer projection and then by binary suffix data. At depth  $L$ , let

$$P_L(M) = \sum_{j=0}^{L-1} \varphi(M+1+j) 2^{L-1-j}$$

be the length- $L$  window numerator underlying Definition 5.3, and define

$$W_{t,L} = P_L(2H_t) - P_L(H_t), \quad \eta_{t,L} = W_{t,L} \bmod 2^L,$$

with the residue chosen in  $[0, 2^L)$ . The finite residue certificate is the asymmetric safe interval

$$H_t + L + 2 < \eta_{t,L} < 2^L - (2H_t + L + 2).$$

The two unequal margins are the tail bounds at the two cone vertices  $H_t$  and  $2H_t$ ; replacing them by a symmetric slogan would lose part of the checked statement.

**Corollary C.9** (finite residue projection). *For every  $t, L \in \mathbb{N}$ , the diagonal projection certificate is equivalent to*

$$H_t + L + 2 < \eta_{t,L} < 2^L - (2H_t + L + 2),$$

*and either condition forces full-target avoidance at height  $t$ . Consequently,*

$$\forall t_0 \in \mathbb{N} \exists t \geq t_0 \exists L \in \mathbb{N}, \quad H_t + L + 2 < \eta_{t,L} < 2^L - (2H_t + L + 2)$$

*implies that  $S$  is irrational.*

Formalised: [residue-projection equivalence](#) and [unbounded projection criterion](#). The finite equivalence is only between the residue inequality and the projection certificate. Neither condition is asserted to be necessary for full-target avoidance or for irrationality. The stronger adjacent-suffix supplies formalised in the same module imply the displayed cofinal supply; they are not additional conclusions of one finite certificate. Finite instances of this implication are proved, but no unbounded projection or suffix supply is established.

**Primality certificates at changing-lcm scales through  $t = 64$ .** A finite family of Lucas certificates proves the primality facts required by the diagonal-pincer certificates at every changing-lcm scale through  $t = 64$ ; for example, [representative Lucas primality certificate](#). This is a finite theorem family, not an unbounded certificate supply or an irrationality proof.

**The diagonal certificate at  $t = 64$ .** Using those primality facts, Lean checks the individual diagonal-pincer certificate at  $t = 64$ , [scale-64 diagonal certificate](#). This is one finite certificate, not an unbounded supply.

### C.3 Candidate cofinal inputs and no-go boundaries

The three-rank condition below is a candidate cofinal sufficient condition, while its affine form exposes a precise obstruction to obtaining that condition from uniform correction bounds alone. The remaining results are independent filters, not consecutive proof steps: the fresh-prime decomposition and one-bit lift expose exact correction terms that an unbounded argument would have to control, and the parity, prime-adjunction, scalar-localisation, and square-CRT results rule out four natural shortcuts. No no-go result uses the preceding filter as a hypothesis.

**A three-rank odd window.** At an analytically admissible odd power-of-two rank, centrality at any one of the canonical odd rank or its next two base-four successors yields the required adjacent-suffix gap condition, [three-rank suffix-gap implication](#). Thus a cofinal three-rank band supply implies irrationality, [cofinal three-rank criterion](#); that supply is not proved.

**Affine form of the three-rank condition.** With the stated short-correction bounds, the three central-band predicates are equivalent to one three-scale affine escape condition, [three-scale affine equivalence](#). A zero-start affine orbit shows that a uniform correction envelope alone does not force escape at any finite number of ranks, [zero-orbit obstruction](#).

**Fresh-prime deficit decomposition.** The literal foreign channel also has an endpoint form suited to curvature tests. Retain only the prime support already present at  $H$  in the two endpoints  $H + s$  and  $2H + s$ , and write the resulting discrepancy from the actual totient as an endpoint deficit. Each such deficit is nonnegative. The actual diagonal increment then equals the old-prime increment plus the lower deficit minus the upper deficit. The second and five-point curvature operators preserve this exact split; the latter has coefficient row  $[-8, 4, 2, 1, 1]$ . Consequently, the old curvature margin loses at most the stated adverse weighted deficit.

**Proposition C.10** (fresh-prime deficit decomposition). *The endpoint fresh deficits are nonnegative, give an exact decomposition of the foreign channel, and bound the loss from the old-prime curvature margins.*

Formalised: [endpoint-deficit nonnegativity](#), [foreign-increment decomposition](#), and [adverse-deficit curvature bound](#). This supplies the exact correction term for the finite evaluations. It supplies neither a cofinal deficit bound nor an irrationality criterion.

**One-bit lifting.** The power-of-two signed-margin condition admits a separate exact arithmetic reduction. If a signed difference is already divisible by  $2^k$ , lifting its congruence to  $2^{k+1}$  is equivalent to evenness of the quotient after division by  $2^k$ . Failure of the lift gives the unique translated residue class. In particular, a displayed difference  $P - N = 16c$  agrees modulo 32 exactly when  $c$  is even; otherwise the residue is the offset class by 16.

**Proposition C.11** (one-bit signed-margin lift). *The fifth-bit question after fourth-bit agreement is exactly one cofactor parity test, with the complementary offset class characterising failure.*

Formalised: [one-bit divisibility lift](#), [mod-32 cofactor criterion](#), and [offset-16 complement](#). The centred-state formulation identifies the signed-margin interval condition with failure of this next lift, subject to its stated correction and margin budget: [centred-lift equivalence](#) and [centred mod-32 equivalence](#). This is a finite arithmetic equivalence. It does not supply the required cofactors at unbounded scales and gives no irrationality conclusion.



**Parity is not a margin certificate.** The preceding lift does not settle the reduced signed-margin interval condition. For a positive even modulus  $m$  and integral radius parameter  $a$  with  $16 < a < 8m - 16$ , each parity class contains both a residue inside the interval and one outside it. The same statement applies to a positive power-of-two modulus. Thus even a fifth-bit computation, by itself, cannot decide the required margin condition.

**Proposition C.12** (parity does not decide the reduced margin condition). *Let  $m, a \in \mathbb{Z}$ , with  $m > 0$  even and*

$$16 < a < 8m - 16,$$

*and put*

$$\mathcal{M}_{m,a} = \{y \in \mathbb{Z} : \exists k \in \mathbb{Z}, 16|y - km| < a\}.$$

*Each parity class contains both a member and a nonmember of  $\mathcal{M}_{m,a}$ . In particular, the conclusion holds for  $m = 2^j$  whenever  $j \geq 1$ .*

Formalised: [parity obstruction](#) and [power-of-two parity obstruction](#). The result is structural and proves no signed-margin certificate at any particular scale; it excludes parity as a sufficient standalone criterion.

**Prime-adjunction no-go theorem.** One tempting way to seek an unbounded obstruction is to compare the full target at  $H$ ,  $pH$ ,  $qH$ , and  $pqH$ . The following affine transport law rules out this route before any projection is introduced.

**Proposition C.13** (prime-adjunction transport collapse). *For every positive  $H$  and  $k$ , the actual diagonal satisfies*

$$D_{kH} = Q_k(H)D_H + Z_k(H), \quad Q_k(H) \in \mathbb{N}, \quad Z_k(H) \in \mathbb{Z}.$$

*Hence integrality at  $H$  propagates to every multiple, and the four target hits at the prime-adjunction diamond are equivalent to the single hit at its root. The corresponding transport of the complete foreign defect has zero diamond curvature: the two transport compositions around the diamond agree exactly.*

Formalised: [the prime-adjunction diamond collapse](#) and [the foreign-correction flatness](#). Thus explicit-shadow fibres alone cannot create a transport obstruction. A future positive route would need a separately defined projection together with a theorem controlling what that projection discards. This is a proved obstruction to one strategy, not progress on the unbounded-supply proposition.

**Proposition C.14** (scalar-localisation obstruction). *Let  $x \in \mathbb{Q}$  and  $c \in \mathbb{Z}$ . If  $H \mid \text{den}(x)$  and the reduced denominator of  $cx$  divides  $H$ , then*

$$\frac{\text{den}(x)}{H} \mid |c|.$$

Formalised: [complementary-denominator divisibility](#) and [scalar-localisation identity](#). Thus a scalar localisation can move a complementary denominator factor into its coefficient, but cannot erase it. This rules out a scalar-denominator shortcut of that form. It neither proves full-target avoidance nor supplies any certificates, so it gives no irrationality criterion for  $S$ .

**Proposition C.15** (square-CRT correction suppression). *Let  $E$  be a finite family indexed by distinct primes  $p_i$ , with prescribed anchors  $A_i$  and residues  $a_i$ . There is a bounded common base  $n$  such that, for each  $i \in E$  and each shift  $h$  with  $p_i \nmid a_i + h$ , one has*

$$n = A_i + p_i(a_i + p_it) \quad \text{and} \quad \varphi(n + p_ih - A_i) = (p_i - 1)\varphi(a_i + p_it + h)$$

*for a suitable  $t \in \mathbb{N}$ . In particular, if every  $p_i$  exceeds a fixed horizon  $J$ , the choice  $a_i = 0$  works simultaneously for all  $1 \leq h \leq J$ .*



Formalised: [finite clean totient family](#) and [clean horizon family](#). The finite mechanism does not force a cube coefficient to be nonzero: there is a checked clean two-step cube with both displayed coefficients zero ([vanishing clean block](#)) and a separate checked clean cube with second coefficient  $-4$  ([nonzero clean block](#)). Thus the result gives neither an unbounded certificate supply nor an irrationality criterion for  $S$ .

#### C.4 Unconditional dyadic rank and carry anti-compression

The determinant construction in this route is unconditional. At every level, a CRT construction prescribes one target prime channel and fresh off-target prime divisors; Dirichlet's theorem supplies the required primes in the resulting progressions. After the exact powers of two are divided from the columns, the evaluation matrix is the identity modulo two. This proves a nonzero separated minor at every level rather than merely postulating one.

**Theorem C.16** (infinite dyadic totient-kernel rank). *For every  $e \geq 0$ , the canonical level- $e$  dyadic totient-kernel family is linearly independent over  $\mathbb{Q}$  and its span has dimension  $2^e + 1$ . Consequently, the span of all sections*

$$n \mapsto \varphi(2^j n + r), \quad j \geq 0, \quad 0 \leq r < 2^j,$$

*is not finite-dimensional over  $\mathbb{Q}$ .*

Formalised by the all-level minor construction [CRT–Dirichlet separated minors](#), the independence theorem [canonical kernel independence](#), the exact rank formula [exact canonical rank](#), and [infinite full-kernel dimension](#). The theorem concerns the coefficient kernel. It does not imply that  $S$  is irrational.

**Relation to prior work.** The qualitative conclusion is not new. Finite-dimensionality of the span of all sections is 2-regularity of the sequence in the sense of Allouche and Shallit [11], and Coons proved in 2010 that  $\varphi$  is not  $k$ -regular for any  $k \geq 2$ , by counting poles of  $\zeta(s-1)/\zeta(s)$  against the meromorphic continuation available to the Dirichlet series of a regular sequence [12]. In particular the infinite-dimensionality of the full kernel span follows from Coons's theorem. What Theorem C.16 adds is quantitative and constructive: the exact canonical rank  $2^e + 1$  at every level, with an explicit separated minor, proved by elementary CRT–Dirichlet arithmetic that the formal source checks without analytic continuation. Proposition C.17 uses exactly this level-by-level form.

**Proposition C.17** (finite-level carry anti-compression). *If the totient coefficient series were rational, there would be a positive-multiplier tempered integral carry orbit whose canonical carry-section span has dimension at least  $2^e - 1$  for every  $e$ .*

Formalised: [the carry-kernel rank lower bound](#) and [the unconditional unbounded-rank consequence of rationality](#). No theorem bounds the dyadic section rank of every rationality-supplied tempered carry. The proposition therefore identifies an infinite rationality-side barrier, not an irrationality criterion for  $S$ .

*Remark* (rank alone cannot decide rationality). No converse dimension bound is available at this generality: the same coefficient class contains rational series whose tempered orbits have infinite-dimensional section span. Choose Boolean values  $a_N \in \{0, 1\}$  with  $a_0 = a_1 = 1$  whose dyadic sections satisfy no nontrivial rational linear relation (a routine diagonal choice on sparse index blocks); set  $u_N = N + a_N$  and  $\gamma(N+1) = 2u_N - u_{N+1}$ . Then  $0 \leq \gamma(n) \leq n$ , the orbit  $u$  is tempered, and Theorem B.1 evaluates  $\sum_{n \geq 1} \gamma(n)2^{-n} = u_0 = 1 \in \mathbb{Q}$ ; yet the sections of  $u$  span an infinite-dimensional space, because the sections of  $N \mapsto N$  span only two dimensions. This elementary observation is not formalised in the source. It fixes the shape of the missing ingredient: a route through Proposition C.17 must couple carry-section rank to the arithmetic size of a totient carry, since dimension alone is compatible with rationality.

**Proposition C.18** (rectangular determinant expansion). *For matrices  $M : \iota \times \kappa \rightarrow R$  and  $N : \kappa \times \iota \rightarrow R$  over a commutative ring, the determinant of  $MN$  is the sum over all maps  $p : \iota \rightarrow \kappa$  of the determinant of the corresponding selected-column matrix times the diagonal product  $\prod_i N_{p(i),i}$ .*

This is the [rectangular determinant expansion](#).

**Proposition C.19** (dominant dyadic term). *If one selected integer numerator is odd and its dyadic exponent is strictly larger than every other selected exponent, then the common numerator after dyadic denominator clearing is odd and hence nonzero.*

Formalised by [dominant dyadic parity](#), and [dominant dyadic nonvanishing](#). These are finite algebraic inputs to a signed-moment route. They do not show that any actual totient Hankel determinant is nonzero, and they give no irrationality criterion for  $S$ .

The formal proof also exposes the construction-independent [separated-minor rank criterion](#). The two generic algebraic propositions above remain useful as possible inputs to other determinant routes, but the totient-kernel minor itself is supplied at every level by Theorem C.16.

**Proposition C.20** (even-residue reduction). *Every even residue channel in the canonical dyadic totient-kernel family is exactly a scalar multiple of its odd-core channel.*

Formalised as the [even-residue reduction](#). This removes duplicate channels before the separated minor is constructed.

**Proposition C.21** (compressed-adjoint contradiction). *A compressed-adjoint certificate whose nonzero multiple is strictly smaller in absolute value than its positive modulus is inconsistent.*

Formalised as the [compressed-adjoint contradiction](#). This is a conditional finite contradiction, not a source of certificates at every level.

**Proposition C.22** (residual-gauge obstruction for monomial minors). *Let a monomial matrix have a common nonzero residual weight in each column. The residual weights act by a right diagonal gauge, and therefore preserve determinant nonvanishing. If one exponent is 1, the inverse-phase locked gauge makes that row identically 1 while preserving a nonzero determinant whenever the coefficient-side determinant is nonzero. If the residual is allowed to depend on both row and column, it reconstructs a consecutive-power matrix whose zeroth row is identically 1.*

Formalised: [residual-gauge determinant equivalence](#), [locked-reconstruction nonvanishing](#), and [reconstructed zero row](#). Thus a residual-blind rank, determinant, or conditioning criterion alone cannot exclude the target. The proposition does not exclude a criterion that also uses an arithmetic coupling identity, and it gives no irrationality criterion for  $S$ .

**Appendix consequence.** Nothing in this appendix strengthens the fixed-parameter completeness of Proposition 5.4 into an unbounded supply. Its remaining condition is therefore the still-open quantifier in Theorem 5.6:

$$\forall h \geq 1 \ \forall N_0 \ \exists N \geq N_0 \ \exists L \geq 1, \quad \text{Sep}(h, N, L).$$

The one-scale mechanisms and conditional implications in Appendix C.2, the candidate inputs and no-go boundaries in Appendix C.3, and the conditional rank routes in Appendix C.4 delimit possible approaches to this statement. None provides the required certificates after every cutoff.

## D Lambert, probability, and finite-algebraic complements

This appendix supplies the extensions cited from the Lambert-series identities in Section 3 and from the coprime-pair representation after Proposition 5.2. Its new results are the periodic-weight classifications, finite denominator obstructions, the Möbius-square and squared Lambert identities, and the Stern–Brocot cylinder and continuant laws. These enlarge the two coordinate pictures; they are not additional steps in either open proof.

The three families are parallel, not cumulative. Appendix D.1 proves or classifies weighted Lambert series with coefficient inputs different from the open totient input. Appendix D.2 gives exact identities and finite algebraic boundaries for the squared and Möbius coordinates. Appendix D.3 develops unconditional geometry internal to the coprime-pair probability coordinate. No conclusion about a comparison input transfers irrationality to  $S$ , and none of the three families supplies the cofinal Sep supply theorem.

### D.1 Weighted Lambert comparison families

This subsection changes the coefficient input. Its eventually-periodic theorems settle or classify those comparison series, while the period-four example blocks one residue-blind shortcut. None has the totient weight defining  $S$ , so no irrationality status transfers to Erdős #249.

**Periodic weights.** For  $L_b(\gamma) = \sum_{a \geq 1} \gamma(a)/(b^a - 1)$ , the checked cases are:

**Theorem D.1** (eventually-periodic nonnegative weights). *For every  $b \geq 2$ , an eventually-periodic nonnegative rational weight that is positive at a positive index in its periodic region gives an irrational series.*

Formalised: [eventually-periodic nonnegative criterion](#). This is contained in a broader theorem of Luca–Tachiya [15].

**Theorem D.2** (periodic signed-weight dichotomy). *For a periodic integer weight, the series is irrational or terminating in base  $b$ .*

Formalised: [periodic signed-weight dichotomy](#).

**Proposition D.3** (residue-blind obstruction). *For the period-four weight  $1, 0, -1, 0$ , the divisor coefficient vanishes on  $n \equiv 3 \pmod{4}$ .*

Formalised: [period-four coefficient obstruction](#).

### D.2 Finite algebraic boundaries and squared-Lambert coordinates

This subsection keeps the relevant coordinates but separates three roles: finite denominator-clearing obstructions, the exact Möbius-square identity for  $S$ , and comparison rows in the squared-Lambert calculus. The identity changes coordinates; the finite propositions delimit algebraic methods; and the status of one squared-Lambert row does not pass to another.

**Finite algebraic boundaries.** Put  $a_n = (\varphi * \mu)(n)/n$ . If an integer  $D$  clears these coordinates through  $N$ , the exact obstruction is:

**Proposition D.4** (primitive-coordinate finite-jet obstruction).  *$D$  is divisible by an explicit two-tier primorial  $T_N$ ; hence no fixed  $D$  clears every  $a_n$ .*

Formalised: [two-tier primorial obstruction](#).

For  $d \geq 1$ , define the pointed periodic atom

$$\omega_d(N) = \frac{2^{N \bmod d}}{2^d - 1}.$$

It satisfies the exact carry law

$$2\omega_d(N) - \omega_d(N+1) = \mathbf{1}_{d|N+1}.$$

**Proposition D.5** (finite Mersenne-atom determinant boundary). *Fix  $m \geq 1$ , conductors  $d_j \geq 1$ , indices  $n_{ij} \geq 0$ , and integer column weights  $w_j$ , and set*

$$W_{ij} = w_j \omega_{d_j}(n_{ij}), \quad U_{ij} = 2^{n_{ij} \bmod d_j}.$$

Then

$$\left( \prod_{j=1}^m (2^{d_j} - 1) \right) \det W = \left( \prod_{j=1}^m w_j \right) \det U \in \mathbb{Z}.$$

If the integer on the right is nonzero, then the absolute value of the cleared determinant on the left is at least 1.

Formalised: [Mersenne-atom carry law](#), [determinant-clearing identity](#), and [cleared-determinant lower bound](#). Thus finite Mersenne-atom determinants recover exactly their explicit column denominators; the clearing creates no additional height surplus. These are scoped finite-denominator obstructions, not irrationality criteria for  $S$ .

**Proposition D.6** (the Möbius-square lens).

$$S = \frac{1}{2} + \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2}.$$

Formalised: [Möbius-square identity](#). Writing  $T = \sum_{d \geq 1} \mu(d)/(2^d - 1)^2$ , irrationality of  $S$  is equivalent to irrationality of  $T$ , since  $S = \frac{1}{2} + T$ . The factor  $1/(2^d - 1)^2$  is the probability that  $d$  divides two independent fair-coin waiting times (Section 5.2). The signed constant, the coprime-pair mass, and the tail differences of Section 5.3 are three readings of the same quantity.

**Squared Lambert calculus.** For  $L_2(f) = \sum_{d \geq 1} f(d)/(2^d - 1)^2$ , the denominator is the probability that  $d$  divides two independent geometric waiting times. The formal transfer theorem, [squared Lambert transfer](#), therefore gives a divisor calculus for their gcd.

**Proposition D.7** (squared Lambert identity). *At  $r = 1/2$ , the three rows  $f = \mu, \mathbf{1}, \varphi$  yield respectively*

$$L_2(\mu) = S - \frac{1}{2}, \quad L_2(\mathbf{1}) = \sum_{n \geq 1} \frac{\sigma(n) - \tau(n)}{2^n}, \quad L_2(\varphi) = \sum_{n \geq 1} \frac{P(n) - n}{2^n},$$

where  $P$  is Pillai's gcd-sum function.

Formalised: [constant-weight squared-Lambert identity](#) and [totient gcd-moment identity](#). Postelmans–Van Assche imply irrationality of the middle row: their linear independence of  $1, \zeta_q(1), \zeta_q(2)$  over  $\mathbb{Q}$ , proved whenever  $1/q$  is an integer at least 2, applies at  $q = 1/2$ , and the middle row is  $\zeta_q(2) - \zeta_q(1)$  there [14]. This does not transfer to the Möbius row, which is exactly the open constant  $S - \frac{1}{2}$ .

### D.3 Probability-coordinate geometry: reduced slopes and cylinders

These results live entirely inside the coprime-pair probability coordinate. They normalise reduced directions and quantify Stern–Brocot cylinder and run geometry. They are unconditional geometric consequences and do not imply the open certificate supply.

**Reduced directions and cylinder masses.** The fair-coin pair has a particularly rigid distribution after division by its gcd. For positive coprime  $a, b$ , put

$$\nu(a, b) = \frac{1}{2^{a+b} - 1}, \quad M(a, b) = \frac{1}{(2^a - 1)(2^b - 1)}.$$

The first quantity is the probability that the reduced direction is  $(a, b)$ ; the second is the mass of the Stern–Brocot cylinder rooted at  $(a, b)$ . Let  $M_d(a, b)$  be the total stop mass in the first  $d$  generations below that root, where a node  $(u, v)$  has stop mass  $(2^{u+v} - 1)^{-1}$  and children  $(u + v, v)$ ,  $(u, u + v)$ .

**Proposition D.8** (reduced-direction normalisation). *The reduced directions form a probability distribution,*

$$\sum_{\substack{a, b \geq 1 \\ (a, b) = 1}} \nu(a, b) = 1.$$

This is the [reduced-direction mass one](#).

**Theorem D.9** (Stern–Brocot cylinder recursion and convergence). *At every positive node,*

$$M(a, b) = \frac{1}{2^{a+b} - 1} + M(a + b, b) + M(a, a + b),$$

and, uniformly in  $a, b$  and  $d$ ,

$$0 \leq M(a, b) - M_d(a, b) \leq \left(\frac{2}{3}\right)^d M(a, b).$$

Consequently  $M_d(a, b) \rightarrow M(a, b)$ ; at the root  $M(1, 1) = 1$ .

Formalised: [cylinder mass recursion](#), [depth- \$d\$  remainder bound](#), and [cylinder convergence](#). The recursion is the elementary rational identity obtained after setting  $A = 2^a$ ,  $B = 2^b$ ; the quantitative point is that the two children retain at most two thirds of their parent’s mass.

**Alternating-run stability.** Write an alternating Stern–Brocot word as  $r$  nonempty runs of lengths  $n_i = 1 + e_i$ , with  $e_i \geq 0$ . Starting with  $P_\emptyset = (1, 1)$ , define

$$P_{(n_1, \dots, n_r)} = (n_1 A + B, A) \quad \text{when} \quad P_{(n_2, \dots, n_r)} = (A, B), \quad H(n_1, \dots, n_r) = P_1 + P_2.$$

Thus  $H$  is the arithmetic height of the primitive pair reached by the word. Let  $F_0 = 0, F_1 = 1$ .

**Theorem D.10** (Fibonacci–continuant height stability). *Every  $r$ -run word satisfies*

$$H(1 + e_1, \dots, 1 + e_r) \geq F_{r+3} + F_{r+1} \sum_{i=1}^r e_i,$$

and  $H(1, \dots, 1) = F_{r+3}$ . A defect confined to position  $i \in \{0, \dots, r - 1\}$  has the exact size

$$H(\underbrace{1, \dots, 1}_i, 1 + e, \underbrace{1, \dots, 1}_{r-i-1}) = F_{r+3} + F_{i+2} F_{r-i+1} e.$$

More generally, the checked recurrence gives the full positive multiaffine continuant defect.

Formalised: [Fibonacci height lower bound](#), [multiaffine run-height formula](#), [aggregate defect lower bound](#), [one-site defect identity](#).

**Proposition D.11** (one-run probability mass). *The two orientations with one nonempty Stern–Brocot run have total mass*

$$\frac{1}{2}.$$

This is the [one-run mass](#).

**Proposition D.12** (unit-run denominator exponent). *The natural denominator exponent along the all-unit  $r$ -run is*

$$\sum_{j=0}^{r-1} F_{j+2} = F_{r+3} - 2.$$

This is the [natural denominator exponent](#).

The Lambert and Möbius identities used above are classical [6, 7, 8], and the mediant language goes back to Farey [10]. Postelmans–Van Assche supply the cited irrationality of the constant-weight squared-Lambert row, not these cylinder or run statements. We claim the displayed Lean-checked identities and estimates, not priority for their assembled package. In particular, the exact equality of the Fibonacci height and denominator scales is a boundary, not an irrationality argument: no theorem here proves that a run tail survives denominator clearing.

**Appendix consequence.** The transfer is coordinate extension only. Appendix D.1 enlarges the Mersenne–Lambert comparison family in Section 3; Appendix D.2 supplies the Möbius-square and gcd-moment coordinates; and Appendix D.3 refines the coprime-pair representation in Proposition 5.2. These are three independent coordinate extensions. None implies the cofinal **Sep** supply for  $S$  or a proof of  $1/2 \in \mathcal{A}$ . Those, rather than any result internal to this appendix, remain the missing statements identified in the main text.

## E Guide to the formal sources

The paper is meant to be read as mathematics. The blue mathematical phrases following formalised statements link to exact checked theorems in the pinned public source revision. All such links use one published source pin; exact declaration names remain in the machine-readable audit layer.

For a first reading, the principal landmarks are Theorem 5.1 for the totient denominator exclusion; Proposition 5.4 and Theorems 5.5–5.7 for the exact certificate equivalences and diagonal form; and Theorems 4.11 and 4.14 for the two half-value classifications. Theorem 4.1 records the classical full-support result, and Section 6 states the structure the reductions share. For the strongest independent structure beyond the two open reductions, see Theorem C.16, the sublogarithmic zero-window corollary in the main #257 results, and Theorems D.9 and D.10. The public repository contains a complete searchable cross-reference for readers who need declaration-level detail.



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